GPSMamba: Global-Perception Spatial-Semantic Enhanced Mamba for Hyperspectral Image Classification

Lianhui Liang, *Member, IEEE*, Chenyang Meng, Shuai Yuan, Zhi Li, Bing Tu, *Senior Member, IEEE*, Jun Li, *Fellow, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*—Hyperspectral image classification (HSIC) is a core task in remote sensing. Traditional convolutional neural networks (CNNs) are constrained by limited local receptive fields and struggle to capture long-range dependencies within hyperspectral images (HSIs). Although Mamba-based HSIC models can effectively model long-range dependencies, flattening 2-D HSI patches into 1-D sequences results in spatial information fragmentation, which creates a pervasive problem of lost spatial continuity and directional cues. To address these limitations, this article proposes a novel global-perception spatial-semantic enhanced mamba network (GPSMamba) for HSI feature extraction. Specifically, the dynamic mixed convolution (DMC) module computes affinity matrices between feature tokens and regional centers via global information embedding (GIE), then aggregates these matrices to generate spatially variant dynamic convolution kernels that jointly model the global context and local features of HSI. Considering the intrinsic spatial–spectral structure of HSI, the direction-aware mamba (DAM) module maintains intact spatial layouts via continuous 2-D scanning and direction-aware updates. Meanwhile, the decoupled spectral attention (DSA) module decouples HSI features along spatial and channel axes, producing dedicated spatial attention matrices and channel attention vectors, respectively. The channel branch captures interband spectral dependencies of HSI, while the spatial branch models intraband pixel correlations, which enhances spectral characterization and spatial structure perception of hyperspectral data. Extensive experiments verify that the proposed approach achieves superior performance against state-of-the-art HSIC methods.**

***Index Terms*—Decoupled spectral attention (DSA), dynamic mixed convolution (DMC), hyperspectral image classification (HSIC), Mamba.**

Received 23 June 2026; revised 23 July 2026; accepted 10 August 2026. Date of publication 12 August 2026; date of current version 25 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62501177 and Grant T2225019; in part by the Innovation Project of Guangxi Graduate Education under Grant YCSW2026131; in part by Guangxi Science and Technology Major Program under Grant AA23073019; in part by Guangxi Natural Science Foundation under Grant 2025JJB170042; in part by the Opening Fund of the University Engineering Research Center of "Satellite +" Space AI Intelligent Governance of Natural Resources, Guangxi, under Grant WXKJAI20260004; in part by the Opening Fund of the Key Laboratory for Coastal Marine Eco-Environment Processes and Carbon Sinks of Hainan Province underr Grant 2025SYKFKT05; in part by the Project for Enhancing Young and Middleaged Teacher's Research Basis Ability in Colleges of Guangxi under Grant 2025KY0050; and in part by the Project of Guangxi Key Research and Development Program under Grant GuikeAB25069412. *(Corresponding author: Zhi Li.)*

Lianhui Liang is with the School of Electrical Engineering, Guangxi University, Nanning 530004, China, and also with the School of Geography and Planning, Nanning Normal University, Nanning 530001, China (e-mail: Lianglh0308@gxu.edu.cn).

Chenyang Meng and Shuai Yuan are with the School of Electrical Engineering, Guangxi University, Nanning 530004, China (e-mail: 2466827241@qq.com; yuanshuai206@126.com).

Zhi Li is with the School of Geography and Planning, Nanning Normal University, Nanning 530001, China (e-mail: lizhi@nnnu.edu.cn).

Bing Tu is with the Institute of Optics and Electronics, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: tubing@nuist.edu.cn).

Jun Li is with Hubei Key Laboratory of Intelligent Geo-Information Processing, School of Computer Science, China University of Geosciences, Wuhan 430078, China (e-mail: lijuncug@cug.edu.cn).

Antonio Plaza is with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politécnica, University of Extremadura, 10071 Caceres, Spain (e-mail: aplaza@unex.es).

Digital Object Identifier 10.1109/TGRS.2026.3723212

I. Introduction

A HYPERSPECTRAL image (HSI) contains hundreds of contiguous spectral bands, which simultaneously capture abundant spectral characteristics and spatial details of ground features, enabling fine ground-object classification [1], [2], [3], [4], [5]. Thus, hyperspectral image (HSI) has been widely applied in many critical scenarios such as mineral identification [6], military missions [7], [8], and carbon stock estimation [9], providing reliable data support for pixel-level identification and refined analysis of ground objects. As the fundamental task of HSI processing, HSI classification (HSIC) aims to achieve accurate discrimination of ground-object categories, and its performance directly affects the reliability of downstream remote sensing applications.

In deep learning-based HSIC, extracting joint global and local spatial–spectral features is the core technique for model feature learning and generalization in complex scenes.

To improve this performance, numerous HSIC methods have been proposed by researchers. Early classification approaches are dominated by traditional machine-learning methods. The maximum likelihood classifier (MLC) performs classification based on Gaussian distribution assumptions [10], [11], [12], while the K-nearest-neighbor (KNN) algorithm conducts discrimination using spectral Euclidean distance metrics [13]. Methods including support vector machine (SVM) [14], sparse representation classification (SRC) [15], and linear discriminant analysis (LDA) [16] have further improved classification accuracy. Nevertheless, these methods only employ handcrafted shallow spectral features. They cannot model continuous spectral characteristics and nonlinear spatial correlations, and demonstrate limited capability for handling spectral variations and spectrally similar ground objects.

The rapid advancement of deep learning has provided new solutions for HSIC.

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

Benefiting from local receptive fields and parameter-sharing mechanisms, convolutional neural networks (CNNs) have become the mainstream architecture for HSIC: 1-D CNN extracts spectral features, 2-D CNN captures spatial information, and 3-D CNN learns joint spatial–spectral features of HSI. A variety of optimized CNN architectures have been proposed to enhance feature extraction performance. Bruni et al. [17] introduced balance constraints for 3-D convolution kernels to alleviate spatial–spectral dimensional imbalance and improve classification stability. Liu et al. [18] proposed multiscale large kernel convolutional neural network (MSLKCNN) with asymmetric multiscale large kernels to expand receptive fields and extract local details and global dependencies simultaneously. Wang et al. [19] designed adaptive rectangular convolution (ARConv) to adaptively adjust kernel size and sampling points to match the spatial distribution of HSI data. Ma et al. [20] presented the lightweight lightweight spatial–spectral deformable convolutional neural network (LS2DCNet) equipped with spatial–spectral deformable convolution (S2DConv) to achieve spatial geometric adaptation and automatic spectral band selection. On this basis, Zhang et al. [21] designed a three-branch adaptive convolution module triple-branch convolutional adaptive spectral–spatial coupling module (TCASSCM) to mine complementary multiscale spatial–spectral features. Chen et al. [22] proposed multichannel hybrid 2D–3D convolutional neural network (MH-2D-3D-CNN), which utilizes multiple feature branches to effectively fuse spatial and spectral features. Nevertheless, existing CNNs cannot adaptively model long-range dependencies while preserving the local inductive bias of convolution, which limits model performance on high-resolution HSI inputs and readily induces feature over-smoothing.

To compensate for the limitations of CNNs, the transformer architecture was introduced into HSIC to achieve global correlation modeling via the self-attention mechanism that adaptively captures pixel correlations. Hong et al. [23] designed SpectralFormer, redefining HSIC from a sequential perspective and constructing the first pure transformer baseline for this task. Jia et al. [24] developed CenterFormer, a central spectral–spatial attention transformer network that strengthens central pixel feature guidance and precisely models global spectral–spatial dependencies. Zhang et al. [25] further proposed CentralGraphFormer, combining central supervision and graph attention, enhancing discriminative feature aggregation in central neighboring regions. To optimize the spectral–spatial feature fusion paradigm, Zhao et al. [26] presented the self-cross augmented transformer (SCAET). This model employs a dual-branch structure to decouple and extract spatial and spectral features, and fuses features via self-cross attention to avoid early fusion loss. On this basis, Jiang et al. [27] proposed the central pixel-guided dual-path multihead attention fusion transformer (CPFormer), which builds central pixel-guided dual-path attention to boost ground-object recognition capability. Wang et al. [28] proposed cross-scene diffusion-enhanced uncertainty attention network (CDUAN), a diffusion-enhanced uncertainty attention network, which integrates diffusion models and uncertainty-aware attention to achieve cross-domain feature alignment. Despite the progress of transformer-based methods, most fuse spatial and channel information or only focus on channel-wise spectral correlations. This readily causes the loss of spectral band dependencies and semantic ambiguity for similar ground-object features, further restricting model performance in small-target and fine-grained classification tasks.

In recent years, state-space model (SSM)-based Mamba has provided a new approach to model long-range spatial–spectral dependencies for HSIC with its excellent long-sequence modeling capability, gradually becoming a mainstream alternative to transformers. Zhu et al. [29] proposed Vision Mamba, pioneering the integration of Mamba into computer vision and laying a foundation for its visual applications. Liu et al. [30] further proposed VMamba to optimize Mamba's structure and refine its basic visual framework. Inspired by these works, He et al. [31] first designed a hybrid Mamba-transformer architecture, which extracts short- and long-range dependencies via multiscale feature extraction and dual-expert collaboration. Ye et al. [32] proposed the direction-aware spatial Mamba network (DASMN), devising a three-direction spatial-enhanced Mamba and combining multiscale CNN to strengthen local features for efficient spectral–spatial extraction. On this basis, Cao et al. [33] embedded the frequency-domain smoothing property of the discrete cosine transform (DCT) into transformer and Mamba, proposing DCT-transformer-Mamba (DTM) to improve the fusion performance of frequency-domain and spatial–spectral features. Liang et al. [34] designed the adaptive rectangular attention Mamba (ARAM), which leverages adaptive convolutions to address Mamba's deficiencies in local feature extraction and long-range feature decay. Xie et al. [35] proposed the Conv-J2SMamba network to strengthen the spatial modeling capability of Mamba and realize the fusion of global and local features. Although Mamba offers a new direction for HSIC, existing Mamba-based classification models fail to fully adapt to visual tasks and HSIC characteristics. Converting 2-D HSI into 1-D sequences leads to core drawbacks, including lost spatial continuity and missing directional information.

To address the weak long-range dependency learning capability of CNNs, spatial fragmentation caused by dimension transformation, and the loss of interband spectral dependencies in HSI, this article proposes a global-perception spatial-semantic enhanced mamba network (GPSMamba) tailored for HSIC. This framework consists of three core components: dynamic mixed convolution (DMC), direction-aware mamba (DAM), and decoupled spectral attention (DSA). In this framework, DMC is first adopted to extract global dependencies and local features of HSI. With the aid of global information embedding (GIE), we calculate the affinity matrix between features and regional centers to obtain spatially adaptive dynamic convolution kernels integrated with long-range dependencies. This approach can dynamically model long-range dependencies while retaining local inductive biases. Second, DAM aims to maintain the spatial continuity of HSI features. It implements spatial sequence scanning (SSS) based on the direction-perception update (DPU) mechanism. This exclusive direction-aware update strategy explicitly encodes directional information during scanning, alleviates spatial frag-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

mentation caused by spatial dimension transformation, and enhances the model's perception of the spatial structures of HSI. Additionally, DSA is introduced to enhance the spectral perception of HSI. By decoupling spatial and channel dimensions and concentrating on spatial attention matrices and channel attention vectors, this approach enables the channel branch to capture spectral dependencies between bands for stronger spectral perception, and the spatial branch to model intraband spatial pixel correlations, which further refines multidimensional spectral perception and the interpretation of HSI spatial structures. After feature fusion, the framework outputs classification results. Compared with existing methods, the proposed GPSMamba achieves superior performance in modeling global spatial–spectral joint features.

The main contributions of this article are summarized as follows.

1) In this article, a novel GPSMamba framework is proposed for HSIC. The framework captures spatial–spectral features of HSI via direction-aware modeling and decoupled attention mechanisms, achieving comprehensive feature extraction for HSI.
2) The DMC is utilized for the first time to generate spatially variant dynamic convolution kernels through computing the affinity between feature tokens and region centers, enabling adaptive encoding of features at different positions in HSI and overcoming the limitations of traditional fixed convolutions.
3) DAM is designed to supplement directional information during SSS while maintaining the continuity of spatial structure. By modeling the spatial features of HSI, this module extracts continuous spatial features and effectively alleviates spatial information fragmentation.
4) By exploiting DSA to decouple HSI data along spatial and spectral dimensions, the spectral dependencies between bands and the pixel correlations within each HSI band are modeled, thus further refining spectral perception and the interpretation of HSI spatial structures.

The rest of this article is organized as follows. Section II briefly introduces the relevant knowledge of the work. Section III elaborates on the proposed method. Section IV presents and analyzes the experimental results. Section V draws conclusions and discusses future work.

II. Related Work

This section briefly introduces the Mamba architecture, squeeze-and-excitation network (SE-Net), attention mechanism, and dynamic convolution, laying the theoretical foundation for the modules and overall framework proposed in Section III.

A. Selective SSM and Mamba (S6)

Mamba originates from SSMs built on control theory, and it was originally designed to resolve the efficiency bottlenecks and performance degradation of traditional transformers when processing long-sequence information—issues stemming from the transformer's quadratic complexity ($O(n^2)$). Early SSMs are linear time-invariant (LTI) systems. Each SSM computes its current state by combining the prior state $x(t) \in \mathbb{R}^n$ and the instantaneous input $u(t) \in \mathbb{R}$; for a system with an n-dimensional state space, the model can be described by the following equations:

$$\begin{cases} \boldsymbol{x}'(t) = \mathbf{A}\boldsymbol{x}(t) + \mathbf{B}\boldsymbol{u}(t) \\ \boldsymbol{y}(t) = \mathbf{C}\boldsymbol{x}(t) \end{cases} \tag{1}$$

where $\boldsymbol{x}'(t)$ is the first-order derivative of the hidden state, $\boldsymbol{u}(t)$ represents the 1-D input at time t, and $\boldsymbol{y}(t)$ denotes the output vector at time t. Matrices $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times n}$, respectively, govern the temporal evolution of hidden states, map inputs to the latent state space, and project latent states to the output space.

However, matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ remain fixed across different inputs, which means SSMs cannot prioritize salient information within input sequences.

With the development of modern control theory and machine learning, long-sequence information has become increasingly prevalent. To address the issues of vanishing or exploding gradients, HiPPO is introduced to handle long-range dependencies. For discretized data, structured SSMs that adopt the zero-order hold (ZOH) technique and incorporate recurrent or convolutional structures internally offer greater advantages in processing such data. The ZOH technique is used to convert matrix $\mathbf{A}$ into the discretized state transition matrix $\overline{\mathbf{A}}$ and matrix $\mathbf{B}$ into the discretized input matrix $\overline{\mathbf{B}}$, and structured SSMs can be expressed by the following equations.

With advances in modern control theory and machine learning, long-sequence data has become ubiquitous. The HiPPO framework was proposed to mitigate vanishing and exploding gradients when modeling long-range dependencies. For discretized data, structured SSMs equipped with the ZOH transformation and internal recurrent or convolutional structures exhibit superior processing performance. The ZOH transform converts matrix $\mathbf{A}$ into the discrete state transition matrix $\overline{\mathbf{A}}$ and matrix $\mathbf{B}$ into the discrete input matrix $\overline{\mathbf{B}}$, and structured SSMs can be expressed via the following equations:

$$\begin{cases} \boldsymbol{x}(t) = \overline{\mathbf{A}}\boldsymbol{x}(t-1) + \overline{\mathbf{B}}\boldsymbol{u}(t) \\ \boldsymbol{y}(t) = \mathbf{C}\boldsymbol{x}(t) \end{cases} \tag{2}$$

where $t-1$ represents the previous discrete time step.

In structured SSMs, the convolution kernel used to accelerate long-sequence training takes the form of a row-concatenated operation, where k denotes the kernel length

$$\overline{\mathbf{K}} = \left(\mathbf{C}\overline{\mathbf{B}} \quad \mathbf{C}\overline{\mathbf{A}\mathbf{B}} \quad \dots \quad \mathbf{C}\overline{\mathbf{A}}^{k}\overline{\mathbf{B}}\right). \tag{3}$$

Subsequently, a key innovation called the selective mechanism was proposed to tackle the limitation that SSMs lack the ability to selectively prioritize important input information, resulting in the selective SSM (S6) that fully exploits the capacities of SSMs. At this stage, an SSM becomes a linear time-varying (LTV) system whose parameters $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ update synchronously as input data changes. Unfortunately, this property prevents precomputation of the convolution kernel given in (3), making parallel training via convolutional structures infeasible for the model.

To speed up training and achieve parallel training, a parallel scanning algorithm inspired by flash attention was devised. It allows the model to process sequential data in parallel while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:32 UTC from IEEE Xplore. Restrictions apply.

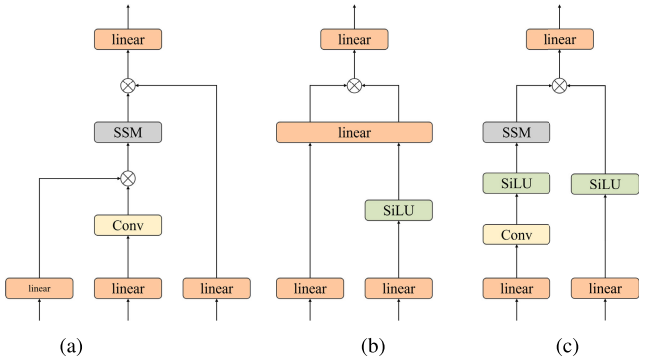


Fig. 1. (a) SSM architecture. (b) Gated MLP. (c) Mamba.

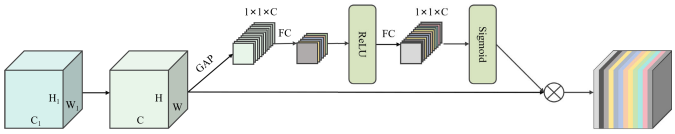


Fig. 2. Architecture of the SE-Net, with squeeze and excitation submodules for adaptive channel feature recalibration.

preserving recurrent computation properties. The complete Mamba block is built by integrating core SSM components with the gated multilayer perceptron (MLP) module. Stacking such blocks alongside normalization and residual connections yields the full Mamba architecture. Fig. 1 shows the fundamental SSM unit [36], [37], the gated MLP [38], and the integrated Mamba block.

B. Squeeze-and-Excitation Networks

SE-Nets can model interdependencies among channels and consist of two core components: the squeeze (compression) component and the excitation component. The detailed structural diagram of SE-Nets is presented in Fig. 2.

The squeeze component compresses global information via channel-wise global average pooling (GAP). The original feature map with a dimension of $H \times W \times C$ is compressed to $1 \times 1 \times C$ through the squeeze operation. The squeeze process can be expressed by the following equation:

$$z_c = F_{\text{sq}}(\mathbf{u}_c) = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} u_c(i, j) \tag{4}$$

where $F_{\text{sq}}(\cdot)$ denotes the squeeze compression function, and $u_c(i, j)$ refers to the feature value at coordinate (i, j) of the cth channel.

The compressed output is fed into the excitation component to capture channel-wise dependencies. This component consists of two fully connected (FC) layers and one rectified linear unit (ReLU) nonlinear activation layer: the first FC layer reduces the number of channels to lower computational complexity, the second FC layer restores the original channel dimension, and the ReLU layer introduces nonlinearity to enhance the model's expressive capacity. The calculation process of the excitation component is defined by the following equation:

$$\mathbf{s} = F_{\text{ex}}(\mathbf{z}, \mathbf{W}) = \sigma(g(\mathbf{z}, \mathbf{W})) = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z})) \tag{5}$$

where $\delta(\cdot)$ denotes the ReLU activation function, $\sigma(\cdot)$ represents the Sigmoid function, $F_{\text{ex}}(\cdot)$ refers to the excitation mapping function, $g(\cdot)$ stands for the composite fully connected transformation function, r is the channel compression ratio, and $\mathbf{z}$ denotes the global channel feature vector. $\mathbf{W}_1 \in \mathbb{R}^{C/r \times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times C/r}$ are learnable weight matrices for dimensionality reduction and dimensional expansion, respectively.

Finally, each feature channel is reweighted to realize dynamic feature recalibration, where $F_{\text{scale}}(\cdot)$ denotes the feature scaling function

$$\tilde{x}_c = F_{\text{scale}}(u_c, s_c) = s_c \cdot u_c \tag{6}$$

where $\tilde{x}_c$ represents the weighted output feature of the cth channel, s_c is the scalar channel attention weight, and $\tilde{\mathbf{x}} = [\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_C]$ denotes the final output feature vector.

C. Attention Mechanism

The attention mechanism is a module used in machine-learning models that mimics the human selective attention process. Its primary objective is to enable the model to focus on the parts of input features most relevant to the current task during information processing and assign corresponding weights to different features, thereby amplifying useful information and suppressing irrelevant content accordingly.

Information is typically processed following this pipeline: "generating queries—calculating matching scores—weighted summation." This mechanism centers on three types of vectors: Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$). All three vectors are obtained through linear transformation, and the corresponding formulation is expressed as follows:

$$\mathbf{Q} = \mathbf{X} \cdot \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X} \cdot \mathbf{W}_K, \quad \mathbf{V} = \mathbf{X} \cdot \mathbf{W}_V \tag{7}$$

where $\mathbf{X}$ denotes the input features, and $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ represent the trainable weight matrices.

The attention score matrix is obtained by computing the similarity between $\mathbf{Q}$ and each $\mathbf{K}$, where the dot product is the commonly used calculation method. Subsequently, the attention score matrix is normalized to generate the attention weights. The process of generating the attention weights can be represented by the following formula:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{8}$$

where Attention$(\cdot)$ is the self-attention function, softmax$(\cdot)$ is the normalization activation function, and d_k is the dimension of the Key vectors. The scaling factor $1/(d_k)^{1/2}$ mitigates gradient vanishing caused by large dot product values in high-dimensional spaces.

D. Dynamic Convolutional Networks

Dynamic convolution is a novel convolutional design that embeds the attention mechanism into convolutional layers to dynamically aggregate multiple parallel convolution kernels. This design improves the model's expressive capacity without increasing the depth or width of the network. Dynamic convolution is built on the perceptron model, where weights

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

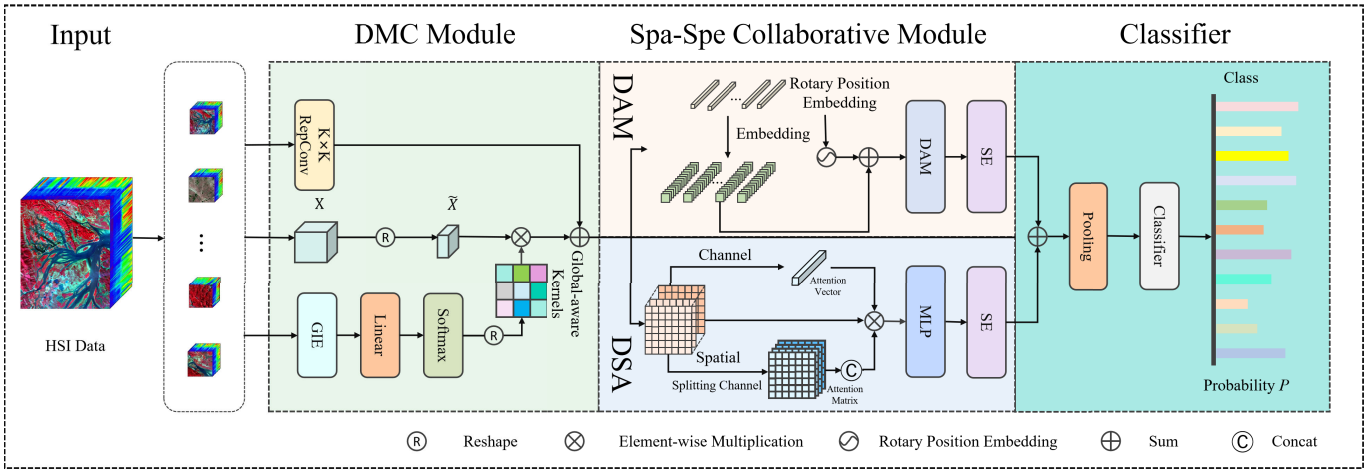


Fig. 3. Overall framework of the proposed GPSMamba for HSIC. The framework consists of the DMC module for aggregating global and local information, the DAM module for modeling spatial structure, and the DSA module for enhancing spectral features.

and biases are obtained via the weighted sum of multiple groups of kernel parameters. These weighting coefficients change along with variations in input data, which means each input sample adopts a unique set of convolutional weights rather than sharing identical parameters across all samples. This approach endows the model with stronger feature representation capability. Extensive experiments have verified that dynamic convolution can effectively boost the performance of convolutional networks. By fusing global and local information and leveraging dynamic convolution, the model can simultaneously balance local inductive biases and long-range dependency modeling capabilities, thus further improving the performance of convolutional networks.

III. Proposed Method

This section first introduces the overall structure of the GPSMamba and then describes its key components in detail. GPSMamba mainly consists of three components: DMC for global–local feature extraction, DAM for spatial feature extraction, and DSA for further enhancing spectral features.

A. Overall Framework

Fig. 3 illustrates the overall architecture of GPSMamba for HSIC, which mainly consists of three core components: DMC, DAM, and DSA. Let the patched HSI cube be denoted as $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, where B denotes the batch dimension, C refers to the spectral channel dimension, and H, W represent the spatial height and width of feature patches, respectively. The patched feature $\mathbf{X}$ is first fed into the DMC module to extract preliminary features that fuse global contextual information and local spatial details

$$\begin{aligned}\mathbf{X}_r &= \text{RepConv}(\mathbf{X})\\ \overline{\mathbf{X}} &= \text{Reshape}(\mathbf{X})\\ \mathbf{X}_k &= \text{Reshape}\left(\text{Softmax}\left(\text{Linear}\left(\overline{\mathbf{X}}\right)\right)\right)\\ \mathbf{X}_c &= \overline{\mathbf{X}} \times \mathbf{X}_k + \mathbf{X}_r\end{aligned} \tag{9}$$

where reparameterized convolution (RepConv) $(\cdot)$ denotes reparameterized convolution, Reshape$(\cdot)$ is the dimension reshaping function, and Linear$(\cdot)$ refers to the FC layer. $\mathbf{X}_r \in \mathbb{R}^{B\times C\times H\times W}$ stands for the local feature extracted via RepConv, $\overline{\mathbf{X}} \in \mathbb{R}^{B\times G\times C/G\times H\times W}$ is the reshaped intermediate feature, $\mathbf{X}_k \in \mathbb{R}^{B\times G\times 1\times K\times K}$ represents the dynamic convolution kernel, and $\mathbf{X}_c \in \mathbb{R}^{B\times C\times H\times W}$ is the output feature of the DMC module. G denotes the number of channel groups, and K stands for the side length of the dynamic convolution kernel with a size of $K \times K$.

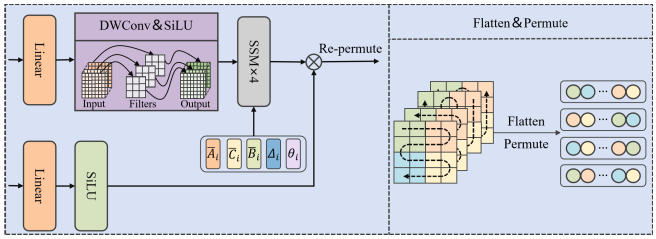


Fig. 4. Schematic of the DAM module structure and the proposed scanning strategy. The left subfigure illustrates the architecture of the developed DAM module, while the right subfigure depicts the scanning paths and sequences of the SSS strategy based on the direction-aware updating mechanism.

Subsequently, the extracted features are split into two parallel branches (DAM and DSA) for further feature extraction to obtain $\mathbf{X}_{\text{spa}}$ and $\mathbf{X}_{\text{spe}}$. Finally, the dual-branch features enhanced by the SE module are fused and fed into the classification head to obtain the prediction results

$$\begin{aligned}\mathbf{X}_{\text{spa}} &= \text{DAM}\left(\text{PosEmb}\left(\mathbf{X}_c\right)\right), \mathbf{X}_{\text{spe}} = \text{DSA}\left(\mathbf{X}_c\right)\\ \mathbf{X}_{\text{fusion}} &= \text{SE}\left(\mathbf{X}_{\text{spa}}\right) + \text{SE}\left(\text{MLP}\left(\mathbf{X}_{\text{spe}}\right)\right)\\ \text{pred} &= \text{Classifier}\left(\text{Pooling}\left(\mathbf{X}_{\text{fusion}}\right)\right)\end{aligned} \tag{10}$$

where position embedding (PosEmb) $(\cdot)$ is the PosEmb, DAM$(\cdot)$ is the DAM module, DSA$(\cdot)$ is the DSA module, SE$(\cdot)$ is the SE attention module, MLP$(\cdot)$ is the MLP, Pooling$(\cdot)$ is the GAP, Classifier$(\cdot)$ is the classification head, $\mathbf{X}_{\text{fusion}}$ is the fused spatial–spectral feature, and pred is the final classification result.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

B. Direction-Aware Mamba

To further extract global–local information and capture spatial features, we propose the DAM module, whose structure is illustrated in Fig. 4. Its core design efficiently adapts Mamba's selective scanning mechanism to 2-D spatial structure scanning and resolves spatial discontinuity and directional information loss during scanning through direction-aware updating.

To adapt to the direction-aware updating of DAM, position encoding is first performed on the features extracted by DMC. The input feature $\mathbf{X}_c$ is flattened to $\mathbf{X}_s \in \mathbb{R}^{B\times L\times C}$ as the input sequence of DAM, where $L = H \times W$ and C is the feature dimension. The feature $\mathbf{X}_s$ is then mapped for dimensional expansion and split into two branches $\mathbf{X}_1 \in \mathbb{R}^{B\times L\times C}$ and $\mathbf{z}_2 \in \mathbb{R}^{B\times L\times C}$ for spatial feature extraction and gating activation, respectively, generating parameters required by Mamba

$$\begin{aligned}\mathbf{X}_1 &= \sigma\left(\mathrm{DWConv}\left(\mathrm{Linear}\left(\mathbf{X}_s\right)\right)\right)\\ \mathbf{z}_2 &= \mathrm{Linear}\left(\mathbf{X}_s\right)\\ \mathbf{B}_i, \mathbf{C}_i, \mathbf{\Delta}_i &= \mathrm{Linear}\left(\mathbf{X}_1\right)\end{aligned} \tag{11}$$

where depthwise convolution (DWConv) $(\cdot)$ denotes depthwise convolution, $\sigma(\cdot)$ represents the Sigmoid activation function, $i \in \{1, 2, 3, 4\}$ refers to the four-directional scanning index, $\mathbf{B}_i$ denotes the input projection matrix, $\mathbf{C}_i$ denotes the output projection matrix, and $\mathbf{\Delta}_i$ represents the time-step parameter.

The branch feature $\mathbf{X}_1$ is temporarily converted into a 2-D feature map. Depthwise convolution is adopted to capture spatial features and supplement local spatial information. The pixels of the 2-D feature map are reordered into a 1-D sequence in a specific order and fed into the SSM selective scanning module for multidirectional scanning of the 1-D sequence.

To update directional information during scanning, the input sequence is reordered along four directions to generate the time steps, input projections, and output projections required by the SSM. Five sets of randomly initialized learnable parameters are introduced into the model to compensate for errors caused by directional variations.

Based on the continuous-time SSM formulated in (1), we introduce the ZOH assumption where the input signal remains constant within a single time interval Δ, that is,

$$\mathbf{x}\left(\tau\right) = \mathbf{x}_l, \quad \tau \in \left[l\Delta,\ (l+1)\,\Delta\right] \tag{12}$$

where l denotes the discrete time index.

We solve the ordinary differential equation via definite integration and obtain the analytical solution of hidden states

$$h_t = e^{\mathbf{A}\Delta} h_{t-1} + \left(\int_0^{\Delta} e^{\mathbf{A}\sigma}\mathbf{B}\,\mathrm{d}\sigma\right) x_{t-1} \tag{13}$$

where $\mathbf{A}$ and $\mathbf{B}$ are the system matrix and input matrix, respectively; h_t stands for the hidden state at time step t.

The universal discrete recursive formula of SSM can be written as

$$\begin{aligned}\overline{\mathbf{A}} &= e^{\mathbf{A}\Delta}\\ \overline{\mathbf{B}} &= \frac{e^{\mathbf{A}\Delta} - \mathbf{I}}{\mathbf{A}}\mathbf{B}\\ h_t &= \overline{\mathbf{A}} h_{t-1} + \overline{\mathbf{B}} x_{t-1}\end{aligned} \tag{14}$$

where $\overline{\mathbf{A}}$ denotes the discrete state transition matrix and $\overline{\mathbf{B}}$ denotes the discrete input matrix.

DAM incorporates the learnable directional correction term $\boldsymbol{\theta}_k$, direction-dependent baseline matrices $\overline{\mathbf{A}}_i$, $\overline{\mathbf{B}}_i$, and time-variant step size $\mathbf{\Delta}_i$. Based on the aforementioned ZOH integral framework, we derive the discrete transformation matrices and directional compensation parameters of DAM. The parameter generation process is defined as follows:

$$\begin{aligned}\overline{\mathbf{A}}_i &= \exp\left(\mathbf{A}_i \cdot \mathbf{\Delta}_i\right)\\ \overline{\mathbf{B}}_i &= \left(\exp\left(\mathbf{A}_i \cdot \mathbf{\Delta}_i\right) - \mathbf{I}\right) / \left(\mathbf{A}_i \cdot \mathbf{\Delta}_i\right) \cdot \mathbf{B}_i\\ \bar{\theta}_{k,i} &= \left(\mathbf{\Delta}_i\mathbf{A}\right)^{-1}\left(\exp\left(\mathbf{\Delta}_i\mathbf{A}\right) - \mathbf{I}\right) \cdot \mathbf{\Delta}_i\boldsymbol{\theta}_k\end{aligned} \tag{15}$$

where $\mathbf{A}_i$ denotes the SSM state transition matrix, $\mathbf{A}$ represents the global baseline state transition matrix, k refers to the index of the learnable parameter group, $\boldsymbol{\theta}_k$ is the base directional parameter, and $\bar{\theta}_{k,i}$ denotes the discretized direction-aware parameter.

After scanning, the sequence is restored to the original 2-D structure via reverse ordering and projected back to the original flattened sequence. Finally, the outputs from the four directions are aggregated to preserve spatial information while conforming to the 1-D computational logic of the SSM. Linear projection back to the input shape yields the output of DAM $\mathbf{X}_{\mathrm{spa}} = \sum_{i=1}^{4} y_i \times z_i$. Taking a single SSM as an example, the whole process is expressed as

$$\begin{aligned}h_t &= \overline{\mathbf{A}}_i h_{t-1} + \left(\overline{\mathbf{B}}_i + \bar{\theta}_{k,i}\right) x_t\\ y_{i,t} &= \mathbf{C}_i h_t + \mathbf{D}_i x_t\\ \mathbf{X}_{\mathrm{spa}} &= \sum_{i=1}^{4} y_i \times z_i\end{aligned} \tag{16}$$

where h_t denotes the SSM hidden state at time step t, h_{t-1} represents the previous hidden state, x_t refers to the input feature element, $y_{i,t}$ stands for the directional output at time step t, $\mathbf{D}_i$ is the output bias matrix, y_i corresponds to the complete directional output, z_i denotes the directional gating feature, and $\sum_{i=1}^{4}$ indicates the summation of results from the four directions.

C. Dynamic Mixed Convolution

Existing dynamic convolutions generate convolution kernels merely using pixels inside local windows and lack global semantic guidance covering the whole image. Deformable convolutions expand receptive fields through offset sampling, yet they undermine the native inductive bias of convolutions. To tackle this drawback, the proposed DMC leverages the GIE to simultaneously extract global long-range dependencies and fine-grained local features within a single forward pass. Accordingly, DMC overcomes the fixed receptive field limitation of traditional CNNs and models global context as well as local details synchronously via spatially adaptive dynamic convolution kernels. The algorithm calculates the affinity matrix between each pixel unit and all regional centers in the feature map to describe the contextual correlation of each pixel. Afterward, all affinity matrices are aggregated to generate spatially variable dynamic convolution kernels that are capable of capturing global dependencies. This approach dynamically

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

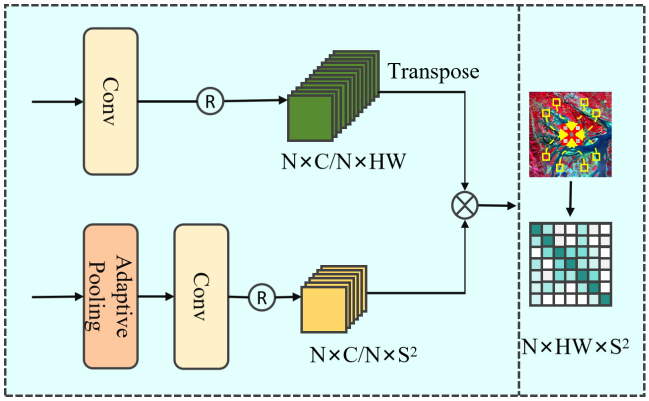


Fig. 5. Structure of the GIE module. The left subfigure presents the generation of variable convolution kernels based on feature affinity, and the right subfigure illustrates the way dynamic convolution embeds global contextual dependencies of HSI.

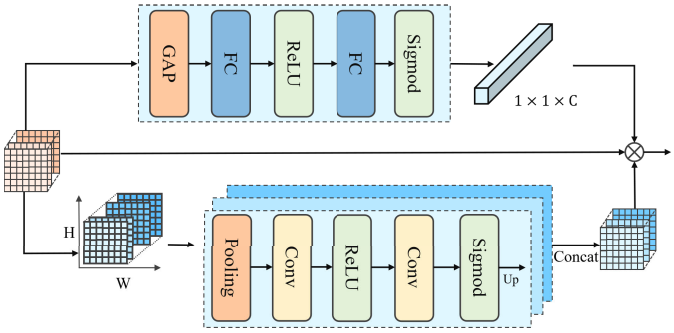


Fig. 6. Structure of the DSA module. Up denotes upsampling.

models long-range dependencies while fully retaining the inherent inductive bias of convolutions.

As illustrated in Fig. 5, the input feature $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$ first passes through two parallel branches: local token generation $\mathbf{T}_l$ and context mixing $\mathbf{T}_g$. Inspired by the multihead attention mechanism, the feature channels are divided into N groups, enabling the model to focus on information from semantic spaces of different dimensions and improve its global modeling capability. In the local token generation branch, the channels of $\mathbf{X}$ are first mapped via convolution, followed by dimensional reshaping to collapse the spatial dimensions $H \times W$ into a 1-D sequence of length HW. The channels are then evenly partitioned to obtain $\mathbf{T}_l \in \mathbb{R}^{B\times N\times C/N\times HW}$. In the context mixing branch, $\mathbf{X}$ is first downsampled to a smaller spatial dimension $S \times S$ through adaptive average pooling and flattened into a 1-D sequence of length S^2, after which the channels are uniformly split to obtain $\mathbf{T}_g \in \mathbb{R}^{B\times N\times C/N\times S^2}$. The context mixing branch that produces $\mathbf{T}_g$ corresponds to the working mechanism of the proposed GIE module. This mechanism aggregates global image semantics merely through simple pooling operations without complicated multihead attention, and generates compact global regional center features to serve as prior information for guiding the generation of dynamic convolution kernels. The whole process can be formulated as follows:

$$\begin{aligned}\mathbf{T}_l &= \text{Reshape}\left(\text{Conv}\left(\mathbf{X}\right)\right)\\ \mathbf{T}_g &= \text{Reshape}\left(\text{Conv}\left(\text{Pooling}\left(\mathbf{X}\right)\right)\right)\end{aligned} \tag{17}$$

where Conv(·) is the standard 2-D convolution, Pooling(·) is the adaptive average pooling, and S is the pooled feature size.

To associate global and local features, the flattened $\mathbf{T}_l$ and $\mathbf{T}_g$ are multiplied to obtain N affinity matrices storing the correlation strength between each token and the corresponding regional center

$$\mathbf{M}_a = \mathbf{T}_l^{\top} \times \mathbf{T}_g \tag{18}$$

where the feature affinity matrix $\mathbf{M}_a \in \mathbb{R}^{N\times HW\times S^2}$ stores the similarity between spatial features and kernel positions.

To associate global information with convolution kernels, the feature information stored in matrix $\mathbf{M}_a$ is mixed via a linear layer, normalized via Softmax, and split into the target kernel shape to obtain convolution kernels containing global features

$$\mathbf{K}_d = \text{Reshape}\left(\text{Softmax}\left(\text{Linear}\left(\mathbf{M}_a\right)\right)\right) \tag{19}$$

where $\mathbf{K}_d \in \mathbb{R}^{B\times N\times 1\times K\times K}$ denotes the dynamic convolution kernel with global dependencies generated by GIE.

To improve training stability, the input features are evenly divided into N groups during feature extraction with residual connections, and D_g is applied to the grouped input features. To preserve the local inductive bias of convolution, a $K \times K$ RepConv is adopted to extract local multiscale features. Finally, the convolution kernel output mapped back to the original dimension is fused with the dilated RepConv output via residual connections to yield the DMC output $\mathbf{X}_c$.

D. Decoupled Spectral Attention

To strengthen the spectral perception of HSI and optimize the understanding of its spatial structures, we introduce DSA. Drawing on SENet, it decouples HSI to extract spectral dependencies separately in spatial and channel dimensions. As shown in Fig. 6, DSA takes the fused feature $\mathbf{X}_c$ as input and extracts spectral features through two independent branches. The channel branch captures interband spectral dependencies, and the spatial branch models intraband spatial pixel correlations.

The channel branch captures spectral dependencies between bands, while the spatial branch models spatial pixel correlations within a single band. First, global spatial information is compressed via GAP to obtain the channel-wise statistical feature $\mathbf{Z}_{\text{spe}}$. Then, two fully connected layers with ReLU activation are adopted for feature compression and reconstruction, and the Sigmoid function normalizes outputs to the interval $(0, 1)$ to generate the channel attention vector $\mathbf{A}_{\text{spe}}$. The calculation process is formulated as

$$\begin{aligned}\mathbf{Z}_{\text{spe}} &= \text{GAP}\left(\mathbf{X}_c\right)\\ \mathbf{A}_{\text{spe}} &= \sigma\left(\mathbf{W}_2\delta\left(\mathbf{W}_1\mathbf{Z}_{\text{spe}}\right)\right)\end{aligned} \tag{20}$$

where $\mathbf{Z}_{\text{spe}}$ denotes the compressed spectral feature obtained after GAP, and $\mathbf{A}_{\text{spe}} \in \mathbb{R}^{1\times 1\times C}$ represents the channel-wise attention weight vector of the DSA.

The spatial branch is designed to mine intraband pixel correlations, and all spectral bands share an identical spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

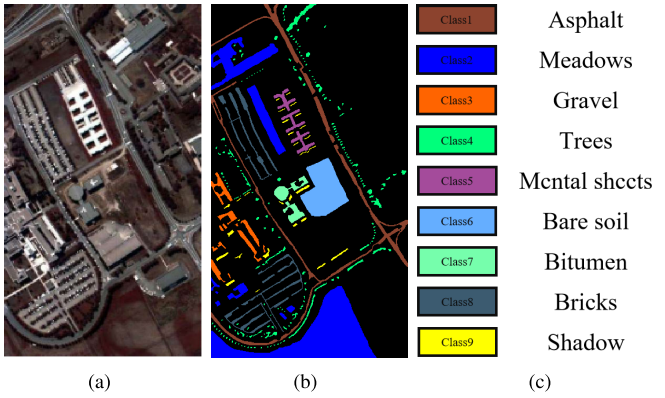


Fig. 7. PU dataset. (a) Pseudo-color map. (b) Ground truth. (c) Legends.

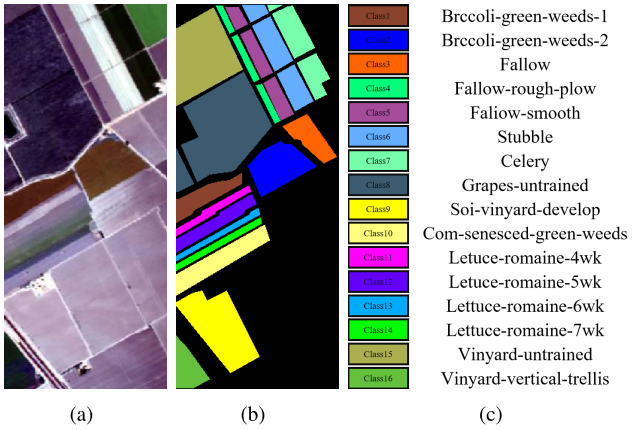


Fig. 8. SA dataset. (a) Pseudo-color map. (b) Ground truth. (c) Legends.

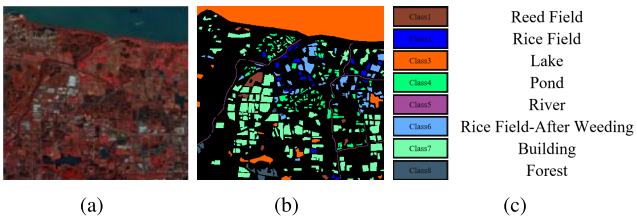


Fig. 9. HZ dataset. (a) Pseudo-color map. (b) Ground truth. (c) Legends.

attention generation pipeline. First, a pooling operation is adopted to compress spatial dimensions, and two convolutional layers with ReLU activation are stacked to extract local spatial textures. Following tensor reshaping and Sigmoid normalization, the single-channel spatial attention mask $\mathbf{A}_{\text{spa-single}}$ is derived, as formulated below:

$$\begin{aligned}\mathbf{Z}_{\text{spa}} &= \text{Pooling}\left(\mathbf{X}_c\right)\\ \mathbf{A}_{\text{spa-single}} &= \sigma\left(\text{Reshape}\left(\text{Conv}\left(\delta\left(\text{Conv}\left(\mathbf{Z}_{\text{spa}}\right)\right)\right)\right)\right)\end{aligned} \tag{21}$$

where $\mathbf{Z}_{\text{spa}}$ denotes the intermediate feature after spatial compression, and $\mathbf{A}_{\text{spa-single}} \in \mathbb{R}^{H\times W\times 1}$ shares the same spatial resolution as the input feature map.

Subsequently, $\mathbf{A}_{\text{spa-single}}$ is processed by the upsampling operation Up(·) to recover the original spatial size, followed by channel-wise concatenation across all spectral bands to construct the complete spatial attention map $\mathbf{A}_{\text{spa}}$. The corresponding formula is given by

$$\mathbf{A}_{\text{spa}} = \text{Concat}\left(\text{Up}\left(\mathbf{A}_{\text{spa-single}}\right)\right), \quad c \in \{1, 2, \ldots, C\} \tag{22}$$

where Concat(·) refers to the tensor concatenation operator along the channel dimension, and $\mathbf{A}_{\text{spa}} \in \mathbb{R}^{H\times W\times C}$ stands for the full spatial attention map of the DSA module.

Finally, the channel attention vector $\mathbf{A}_{\text{spe}}$ and spatial attention map $\mathbf{A}_{\text{spa}}$ are multiplied element-wise with the input feature $\mathbf{X}_c$ simultaneously to achieve dual-dimensional spectral–spatial feature enhancement. The output spectral feature $\mathbf{X}_{\text{spe}}$ of the DSA module is calculated as follows:

$$\mathbf{X}_{\text{spe}} = \mathbf{X}_c \times \mathbf{A}_{\text{spe}} \times \mathbf{A}_{\text{spa}}. \tag{23}$$

IV. Experiments

This section evaluates the proposed GPSMamba on four public hyperspectral datasets, namely Pavia University (PU), Salinas (SA), HuZhou (HZ), and CamPha (CP).

A. Dataset Analysis

This section gives a brief description of four popular datasets used in our experiments.

1) Pavia University: It was collected by the reflective optics system imaging spectrometer (ROSIS) over the campus and surrounding areas of Pavia University (PU), Italy. The original image size is 610 × 340 pixels, with a spectral range of 430–860 nm and 115 original spectral bands. After removing 12 invalid bands affected by low signal-to-noise ratio and water absorption, 103 effective spectral bands are retained. The dataset contains nine land cover types (asphalt, meadow, gravel, trees, bare soil, bricks, metal sheets, and shadows) with a total of 42 776 labeled samples. A false-color composite map and its corresponding ground truth are shown in Fig. 7.

2) Salinas (SA): It was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) over the agricultural region of Salinas Valley, California, USA. The image measures 512 × 217 pixels, with a spectral range of 400–2500 nm and 224 original spectral bands. After removing 20 bands distorted by water absorption, 240 effective spectral bands are retained. The dataset contains 16 land cover types (celery, vineyard, fallow, vineyard soil, and corn) with a total of 54 129 labeled samples. A false-color composite image and its corresponding 16-class ground truth map are presented in Fig. 8.

3) HuZhou (HZ): It was collected by the advanced hyperspectral imager (AHSI) onboard the Gaofen-5 (GF-5) satellite over the Taihu Lake region in Huzhou City, Zhejiang Province, China. The image measures 352 × 379 pixels, with a spectral range of 400–2500 nm and 330 original spectral bands. After removing 28 invalid bands disturbed by atmospheric absorption and instrument noise, 302 effective spectral bands are retained. The dataset contains eight typical land cover types native to the Yangtze River Delta (reed land, paddy field, lake, pond, river, weeded paddy field, building, and forest) with a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

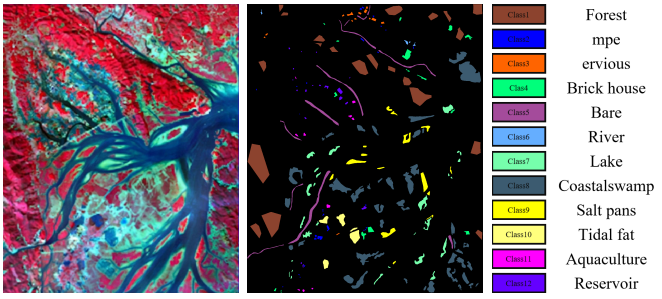


Fig. 10. CP dataset. (a) Pseudo-color map. (b) Ground truth. (c) Legends.

total of 42 179 labeled samples. A false-color composite image and its corresponding ground truth map, which marks eight distinct classes, are illustrated in Fig. 9.

4) CamPha (CP): It was collected by the OHS-1 sensor onboard the Zhuhai-1 satellite over the coastal zone of Cam Pha City, Quang Ninh Province, Vietnam. The image measures 1500 × 1200 pixels, with a spectral range of 400–1000 nm and 32 effective spectral bands. The dataset contains 12 land cover types: eight wetland types (River, Lake, Coastal swamp, Coastal marsh, Salt pans, Tidal flat, Aquaculture, Reservoir) and four non-wetland types (Forest, Impervious, Brick house, Bare). A false-color composite image and its corresponding ground truth map, which marks 12 classes, are illustrated in Fig. 10.

B. Experimental Settings

1) Evaluation Metrics: To ensure experimental fairness, three commonly used evaluation metrics in HSI classification are adopted: Overall Accuracy (OA), Average Accuracy (AA), and Kappa coefficient.

2) Comparison Methods: To fully verify the superiority of the proposed GPSMamba, ten mainstream methods in HSIC are selected for comparison, covering four categories: traditional machine learning (SVM [14], KNN [13]), CNN-based (3DCNN [39], CSCANet [40]), transformer-based [DBCTNet [41], grouped separable convolution (GSC)-VIT [42], DiffFormer [43]], and Mamba-based (3DSS-Mamba [44], IGroupSS-Mamba [45], DBMGNet [46]).

a) SVM: This method classifies by finding the hyperplane that maximizes the margin between samples and the optimal hyperplane in the feature space.

b) KNN: This method classifies by calculating the distance between a new sample and all training samples, and taking the majority or average of the nearest neighbors.

c) 3DCNN: This model directly processes 3-D spatial–spectral data with 3-D convolution kernels to synchronously capture joint spatial–spectral features.

d) CSCANet: This model fuses multispectral spatial complementary information and realizes local–global spatial feature interaction via the cascade spatial cross attention (CSCA) module, improving feature discrimination of different land covers via a dual-branch spatial–spectral feature separation structure.

e) DBCTNet: This model extracts local features via CNN and models global dependencies via transformer for cross-modal fusion of different features.

f) GSC-ViT: This model reduces model parameters and extracts local spatial–spectral features via GSC, and realizes joint local–global spatial feature extraction via the grouped separable self-attention (GSSA) module.

g) DiffFormer: This model proposes the differential multihead self-attention (DMHSA) to highlight differences between spatial and spectral features, and enhances features via the SwiGLU activation function.

h) 3DSS-Mamba: This model introduces a 3-D selective scanning mechanism to model the 3-D structure of HSI and separately capture spatial and spectral long-range dependencies.

i) IGroupSS-Mamba: This model proposes the interval grouped S6 mechanism (IGSM) to divide nonoverlapping groups for redundancy reduction, and cascades spatial–spectral operators to extract global features.

j) DBMGNet: This model fuses the advantages of Mamba and graph convolutional network (GCN), constructing the band selection enhanced bidirectional Mamba (BSEBM) module to model spatial–spectral long-range dependencies and the reparameterized Chebyshev graph convolution (RCGC) module to capture non-Euclidean spatial pixel similarity.

3) Implementation Details: All experiments are conducted on a unified hardware and software platform to ensure fair comparison. All experiments are implemented based on PyTorch, run 20× with 200 epochs each. For different datasets, the training and test sets are divided by percentage. For the PU dataset, the batch size, patch size, learning rate, and training set ratio are set to 64, 9, 1.1×10^{-4}, and 1%, respectively; for the SA dataset, the corresponding parameters are 128, 13, 1.1×10^{-4}, and 1%; for the HZ dataset, the parameters are 64, 7, 1.1×10^{-4}, and 1%; for the CP dataset, the parameters are 128, 11, 1.1×10^{-4}, and 0.2%. All experiments are run on the same hardware platform with the following specifications: Intel[1] Xeon[1] central processing unit (CPU) E5-2696 v4 @ 2.20 GHz, 128 GB random access memory (RAM), 4 NVIDIA Tesla V100-SXM2-16GB GPUs, compute unified device architecture (CUDA) version 12.2, and Ubuntu 24.04 as the operating system.

C. Parameter Analysis

To obtain the optimal combination of hyperparameters and mitigate performance degradation caused by inappropriate parameter settings, this section analyzes the key hyperparameters that affect model performance. We adopt OA as the core evaluation metric to determine the optimal hyperparameter configuration.

1) Analysis of the Patch Size: The patch size determines the receptive field of the model and directly affects the extraction of local spatial–spectral features. All other parameters are fixed in the experiments, and the candidate patch sizes are set as $\{5, 7, 9, 11, 13, 15\}$. As shown in Fig. 11(a), for the PU dataset, OA first increases and then slightly declines with the growth of patch size, which indicates that an excessively large patch size will introduce irrelevant interference factors. For the SA dataset, OA gradually rises and tends to be stable as the patch size expands. For the HZ dataset, OA presents an

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

TABLE I

CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE PU DATASET USING 1% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional		CNN-based Methods		Transformer-based Methods			Mamba-based Methods			GPSMamba
	SVM	KNN	3DCNN	CSCANet	DBCTNet	GSCVIT	DiffFormer	3DSSMamba	IGroupSSMamba	DBMGNet	
class1	89.32 ± 2.87	81.73 ± 2.10	91.40 ± 2.65	97.59 ± 0.97	96.57 ± 3.26	95.73 ± 2.45	85.83 ± 2.55	91.50 ± 3.34	98.40 ± 0.60	98.99 ± 0.55	**99.29 ± 0.47**
class2	97.50 ± 0.82	94.37 ± 1.02	96.58 ± 1.26	**99.98 ± 0.03**	99.25 ± 0.50	98.80 ± 1.17	99.23 ± 0.66	95.80 ± 1.45	99.85 ± 0.09	99.96 ± 0.03	**99.98 ± 0.03**
class3	57.98 ± 8.39	34.62 ± 6.26	25.53 ± 9.60	87.37 ± 6.73	91.45 ± 4.55	85.36 ± 10.95	75.25 ± 6.22	85.20 ± 6.14	92.45 ± 2.88	90.09 ± 3.01	**93.55 ± 1.63**
class4	87.05 ± 2.64	67.85 ± 5.51	86.19 ± 5.22	95.01 ± 1.12	95.63 ± 1.42	94.86 ± 1.84	85.63 ± 2.58	83.91 ± 2.81	90.30 ± 1.45	96.38 ± 0.79	**96.62 ± 0.64**
class5	98.95 ± 0.15	98.86 ± 0.27	98.81 ± 0.90	99.27 ± 0.34	99.92 ± 0.11	99.88 ± 0.19	98.95 ± 1.92	98.05 ± 2.11	98.76 ± 0.52	**99.98 ± 0.05**	99.78 ± 0.60
class6	69.82 ± 4.89	34.97 ± 3.03	31.98 ± 5.24	99.84 ± 0.22	99.39 ± 0.60	97.91 ± 1.12	74.81 ± 3.23	81.22 ± 8.05	99.31 ± 0.61	98.97 ± 0.58	**99.72 ± 0.33**
class7	66.26 ± 7.95	69.89 ± 6.74	20.61 ± 14.65	**99.77 ± 0.35**	98.62 ± 1.40	89.18 ± 9.06	78.18 ± 12.80	62.65 ± 10.03	99.63 ± 0.42	97.71 ± 1.78	99.04 ± 1.40
class8	83.67 ± 3.05	78.06 ± 5.07	88.70 ± 2.23	95.22 ± 2.42	96.66 ± 2.74	90.53 ± 6.47	72.74 ± 6.28	88.84 ± 3.28	96.19 ± 1.03	96.79 ± 2.07	**97.31 ± 1.64**
class9	**99.73 ± 0.11**	99.70 ± 0.06	93.06 ± 7.85	95.03 ± 1.52	98.27 ± 1.66	98.26 ± 1.43	72.59 ± 7.23	77.46 ± 18.84	99.05 ± 0.46	99.48 ± 0.80	98.24 ± 0.88
OA(%)	88.22 ± 0.52	78.69 ± 0.64	80.91 ± 0.96	98.07 ± 0.33	97.97 ± 0.65	96.29 ± 0.83	88.60 ± 1.22	90.08 ± 1.12	98.14 ± 0.34	98.00 ± 0.21	**98.98 ± 0.18**
AA(%)	83.36 ± 1.10	73.34 ± 0.89	70.32 ± 1.74	96.57 ± 0.74	97.30 ± 0.66	94.50 ± 1.52	82.58 ± 2.70	84.96 ± 2.90	97.10 ± 0.58	97.59 ± 0.51	**98.17 ± 0.35**
Kappa(%)	84.14 ± 0.72	70.81 ± 0.85	73.72 ± 1.39	97.44 ± 0.44	97.31 ± 0.87	95.09 ± 1.09	84.65 ± 1.68	86.85 ± 1.51	97.53 ± 0.45	98.14 ± 0.28	**98.65 ± 0.23**
Training Time	**0.01 ± 0.00**	**0.01 ± 0.00**	11.39 ± 0.34	34.15 ± 1.22	131.21 ± 15.75	20.34 ± 2.35	19.51 ± 0.32	252.79 ± 0.17	133.23 ± 1.26	173.95 ± 5.63	175.75 ± 1.27
Test Time	20.46 ± 0.84	**0.11 ± 0.01**	7.05 ± 0.07	6.08 ± 0.16	10.11 ± 2.39	21.23 ± 2.71	11.49 ± 0.13	52.00 ± 0.08	49.34 ± 1.89	6.48 ± 0.60	24.59 ± 12.99

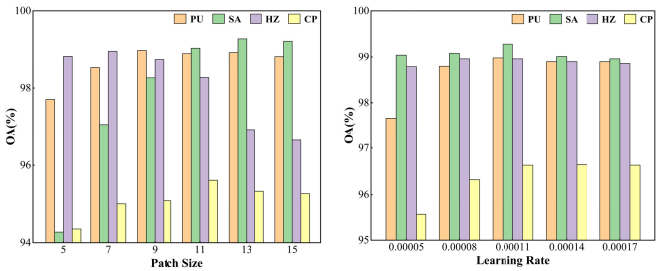


Fig. 11. Analysis of different parameters on OA across PU, SA, HZ, and CP. (a) Effect of patch size. (b) Effect of learning rate.

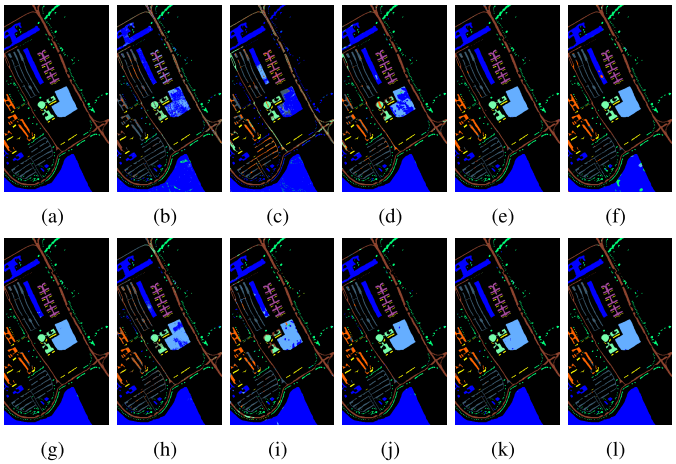


Fig. 12. Classification visualizations for the PU dataset based on 1% labeled training samples per class. (a) Ground truth, (b) SVM, OA = 88.22%, (c) KNN, OA = 78.69%, (d) 3DCNN, OA = 80.91%, (e) CSCANet, OA = 98.07%, (f) DBCTNet, OA = 97.97%, (g) GSC-ViT, OA = 96.29%, (h) DiffFormer, OA = 88.60%, (i) 3DSS-Mamba, OA = 90.08%, (j) IGroupSS-Mamba, OA = 98.14%, (k) DBMGNet, OA = 98.00%, (l) GPSMamba, OA = 98.98%.

overall downward trend when adopting large patch sizes. For the CP dataset, the classification accuracy gradually improves with the increase of patch size, while further expanding the patch size leads to a slow drop, which proves that a moderate patch size can extract sufficient spatial context information. In summary, different datasets require distinct patch sizes to balance feature extraction capability and classification performance.

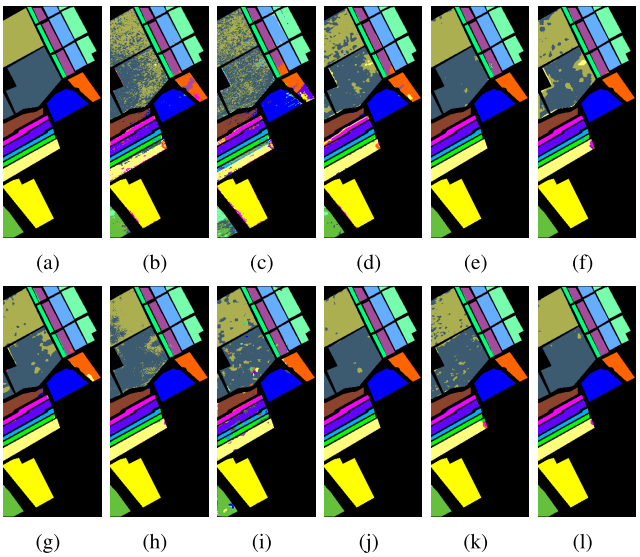


Fig. 13. Classification visualizations for the SA dataset based on 1% labeled training samples per class. (a) Ground truth, (b) SVM, OA = 89.14%, (c) KNN, OA = 83.81%, (d) 3DCNN, OA = 86.90%, (e) CSCANet, OA = 99.21%, (f) DBCTNet, OA = 92.67%, (g) GSC-ViT, OA = 93.44%, (h) DiffFormer, OA = 94.39%, (i) 3DSS-Mamba, OA = 93.84%, (j) IGroupSS-Mamba, OA = 99.14%, (k) DBMGNet, OA = 97.13%, (l) GPSMamba, OA = 99.29%.

2) Analysis of the Learning Rate: The learning rate controls the update step of model parameters and the training convergence speed, which directly determines training stability and classification performance. All other hyperparameters are fixed during experiments, and the candidate learning rates are set as $\{0.00005, 0.00008, 0.00011, 0.00014, 0.00017\}$. As illustrated in Fig. 11(b), the OA of all four datasets rises first and then drops with the increase of the learning rate. When the learning rate increases from 0.00005 to 0.00011, the OA of all datasets steadily grows, demonstrating that moderately raising the learning rate can accelerate model convergence. If the learning rate exceeds 0.00011 and keeps increasing, the classification accuracy of the model generally decreases, and the performance variation of the SA dataset is the most prominent. This reveals that an overly large learning rate easily causes training oscillation and damages the generalization ability of the model. Overall, setting the learning rate to 0.00011 achieves the optimal balance between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

TABLE II

CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE SA DATASET USING 1% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional		CNN-based Methods		Transformer-based Methods			Mamba-based Methods			GPSMamba
	SVM	KNN	3DCNN	CSCANet	DBCTNet	GSCVIT	DiffFormer	3DSSMamba	IGroupSSMamba	DBMGNet	
class1	97.51 ± 1.17	95.40 ± 0.99	94.54 ± 6.29	**100.00 ± 0.00**	99.91 ± 0.16	96.48 ± 4.27	98.79 ± 1.02	98.14 ± 1.81	**100.00 ± 0.00**	99.99 ± 0.03	**100.00 ± 0.00**
class2	99.13 ± 0.59	98.12 ± 0.70	99.38 ± 0.76	99.98 ± 0.06	**100.00 ± 0.00**	98.18 ± 5.12	**100.00 ± 0.00**	99.29 ± 0.43	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
class3	91.76 ± 4.69	87.57 ± 8.50	71.50 ± 5.66	**100.00 ± 0.00**	98.58 ± 2.18	88.21 ± 9.80	99.54 ± 0.29	94.36 ± 4.59	**100.00 ± 0.00**	**100.00 ± 0.00**	99.72 ± 0.83
class4	99.30 ± 0.48	98.59 ± 0.42	97.22 ± 1.99	99.27 ± 0.16	99.57 ± 0.46	97.23 ± 4.24	97.10 ± 0.77	96.77 ± 1.91	**99.70 ± 0.19**	99.63 ± 0.45	98.78 ± 1.25
class5	95.06 ± 3.19	95.68 ± 1.21	93.58 ± 3.56	96.88 ± 1.37	98.23 ± 1.59	95.60 ± 3.49	97.51 ± 1.10	97.69 ± 1.03	**99.75 ± 0.15**	99.22 ± 0.77	99.49 ± 0.54
class6	99.59 ± 0.12	98.67 ± 0.77	99.91 ± 0.22	99.97 ± 0.05	99.95 ± 0.12	99.81 ± 0.34	**100.00 ± 0.00**	99.43 ± 0.44	99.96 ± 0.07	99.99 ± 0.02	99.94 ± 0.13
class7	99.36 ± 0.24	98.81 ± 0.43	99.81 ± 0.14	99.91 ± 0.08	99.90 ± 0.17	99.84 ± 0.45	99.91 ± 0.08	95.73 ± 1.12	99.94 ± 0.07	99.99 ± 0.01	**100.00 ± 0.01**
class8	90.06 ± 1.20	74.45 ± 3.00	80.11 ± 5.45	98.78 ± 0.27	80.82 ± 6.14	88.69 ± 5.17	90.09 ± 1.56	94.02 ± 3.08	**98.82 ± 0.30**	93.07 ± 1.33	98.41 ± 0.72
class9	98.30 ± 0.92	96.15 ± 0.74	98.14 ± 1.29	99.99 ± 0.02	99.90 ± 0.17	99.45 ± 1.08	99.99 ± 0.01	99.82 ± 0.13	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
class10	86.36 ± 4.87	79.27 ± 4.46	83.09 ± 3.75	**99.75 ± 0.10**	95.70 ± 1.14	91.63 ± 6.48	98.06 ± 0.45	92.95 ± 1.91	99.16 ± 0.51	96.91 ± 1.80	98.06 ± 1.82
class11	88.66 ± 4.25	75.75 ± 6.89	43.94 ± 16.71	99.76 ± 0.09	99.31 ± 1.28	84.11 ± 14.97	98.22 ± 0.99	93.11 ± 3.06	98.73 ± 0.95	94.74 ± 4.94	**99.91 ± 0.17**
class12	99.75 ± 0.31	96.27 ± 4.45	98.97 ± 0.98	99.87 ± 0.15	99.87 ± 0.10	98.88 ± 1.00	99.55 ± 1.23	99.01 ± 5.21	99.71 ± 0.80	**100.00 ± 0.00**	99.96 ± 0.05
class13	97.61 ± 0.96	97.88 ± 0.47	99.81 ± 0.20	96.47 ± 1.95	99.75 ± 0.40	98.59 ± 2.05	97.81 ± 1.85	82.45 ± 9.06	96.39 ± 2.12	**99.90 ± 0.20**	99.83 ± 0.16
class14	91.58 ± 2.98	89.20 ± 1.83	98.64 ± 0.96	99.41 ± 0.19	98.60 ± 2.16	97.39 ± 4.34	96.39 ± 2.21	87.07 ± 7.39	96.49 ± 7.52	98.47 ± 1.16	**99.56 ± 0.74**
class15	52.20 ± 4.66	54.62 ± 3.31	68.75 ± 10.01	98.16 ± 0.96	79.13 ± 9.41	85.02 ± 7.00	78.08 ± 3.45	82.83 ± 7.64	97.37 ± 0.77	92.39 ± 1.98	**98.73 ± 0.78**
class16	96.45 ± 2.23	68.81 ± 8.08	80.39 ± 5.25	99.69 ± 0.44	98.58 ± 1.88	94.14 ± 4.06	98.15 ± 1.79	91.74 ± 3.58	99.46 ± 0.16	98.82 ± 0.77	**99.79 ± 0.39**
OA(%)	89.14 ± 0.32	83.81 ± 0.72	86.90 ± 1.26	99.21 ± 0.09	92.67 ± 1.24	93.44 ± 1.33	94.39 ± 0.53	93.84 ± 0.87	99.14 ± 0.24	97.13 ± 0.32	**99.29 ± 0.20**
AA(%)	92.67 ± 0.51	87.83 ± 1.26	87.99 ± 1.29	99.24 ± 0.16	96.74 ± 0.61	94.58 ± 0.171	96.83 ± 0.38	93.46 ± 0.85	99.09 ± 0.55	98.32 ± 0.39	**99.51 ± 0.18**
Kappa(%)	87.87 ± 0.36	81.94 ± 0.81	85.39 ± 1.42	99.12 ± 0.10	91.85 ± 1.38	92.69 ± 0.148	93.74 ± 0.59	93.12 ± 0.97	99.04 ± 0.27	96.80 ± 0.36	**99.21 ± 0.22**
Training Time	0.01 ± 0.01	**0.00 ± 0.00**	21.97 ± 0.50	44.82 ± 0.77	123.38 ± 8.58	32.97 ± 1.72	21.53 ± 0.54	49.87 ± 6.25	183.31 ± 5.86	422.61 ± 164.04	278.11 ± 0.99
Test Time	15.58 ± 0.57	**0.16 ± 0.01**	6.38 ± 0.12	7.43 ± 0.18	5.51 ± 0.26	12.40 ± 0.33	12.98 ± 0.18	7.53 ± 1.77	59.33 ± 4.34	19.49 ± 7.50	13.82 ± 0.07

TABLE III

CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE HZ DATASET USING 1% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional		CNN-based Methods		Transformer-based Methods			Mamba-based Methods			GPSMamba
	SVM	KNN	3DCNN	CSCANet	DBCTNet	GSCVIT	DiffFormer	3DSSMamba	IGroupSSMamba	DBMGNet	
class1	96.00 ± 2.52	83.65 ± 27.85	94.63 ± 1.96	97.68 ± 1.17	99.68 ± 0.25	95.12 ± 8.74	90.98 ± 7.97	65.56 ± 19.83	97.57 ± 1.73	99.35 ± 0.72	**99.46 ± 0.33**
class2	97.42 ± 0.98	90.89 ± 20.29	93.49 ± 2.18	94.22 ± 0.59	96.42 ± 2.65	94.45 ± 3.36	92.97 ± 1.44	25.10 ± 11.11	96.02 ± 1.14	**99.13 ± 0.61**	98.37 ± 1.32
class3	99.71 ± 0.14	98.99 ± 2.20	98.92 ± 0.34	99.84 ± 0.09	99.09 ± 0.56	99.77 ± 0.12	99.31 ± 0.35	98.45 ± 0.40	99.78 ± 0.08	**99.82 ± 0.10**	99.78 ± 0.15
class4	86.29 ± 2.25	82.37 ± 7.06	82.14 ± 5.75	93.48 ± 1.62	95.28 ± 2.58	91.91 ± 2.40	76.19 ± 4.37	61.48 ± 10.46	91.34 ± 1.42	97.17 ± 1.06	**97.34 ± 1.15**
class5	43.97 ± 6.09	34.36 ± 23.48	20.35 ± 8.21	60.05 ± 6.76	73.34 ± 11.29	54.46 ± 3.97	29.43 ± 5.27	24.77 ± 12.12	60.01 ± 4.16	74.65 ± 6.47	**74.90 ± 11.04**
class6	97.18 ± 0.97	95.20 ± 5.85	95.59 ± 1.73	96.89 ± 1.83	98.45 ± 0.85	97.33 ± 2.83	95.57 ± 2.39	80.40 ± 16.28	96.08 ± 1.19	99.52 ± 0.28	**99.65 ± 0.28**
class7	96.30 ± 1.02	87.01 ± 17.72	94.82 ± 2.78	98.97 ± 0.40	98.84 ± 0.43	98.37 ± 1.41	98.48 ± 0.70	90.42 ± 3.70	98.24 ± 0.25	99.56 ± 0.32	**99.80 ± 0.11**
class8	85.35 ± 6.15	77.72 ± 6.87	92.09 ± 4.07	95.01 ± 1.86	98.07 ± 0.74	94.98 ± 3.55	82.21 ± 9.06	38.13 ± 15.08	96.00 ± 1.19	98.17 ± 1.02	**98.62 ± 1.03**
OA(%)	95.90 ± 0.43	93.96 ± 0.36	94.15 ± 0.79	97.72 ± 0.19	98.04 ± 0.28	97.26 ± 0.54	94.97 ± 0.44	86.68 ± 2.11	97.34 ± 0.13	98.90 ± 0.15	**98.97 ± 0.16**
AA(%)	87.78 ± 1.12	82.06 ± 2.97	84.00 ± 1.61	92.02 ± 0.81	94.89 ± 1.17	90.80 ± 1.15	83.14 ± 1.08	60.54 ± 6.25	91.88 ± 0.67	95.92 ± 0.82	**95.99 ± 1.23**
Kappa(%)	93.95 ± 0.62	91.29 ± 1.07	91.38 ± 1.12	96.63 ± 0.28	97.12 ± 0.41	95.96 ± 0.78	92.55 ± 0.63	80.17 ± 3.22	96.08 ± 0.19	98.38 ± 0.22	**98.48 ± 0.24**
Training Time	**0.01 ± 0.00**	0.08 ± 0.25	24.11 ± 0.65	33.36 ± 1.10	169.70 ± 16.90	30.30 ± 0.74	18.44 ± 0.40	37.13 ± 6.20	215.20 ± 61.03	208.33 ± 5.42	148.19 ± 2.60
Test Time	14.63 ± 0.35	**0.35 ± 0.19**	11.30 ± 0.44	5.67 ± 0.12	6.55 ± 0.50	17.69 ± 0.31	10.29 ± 0.10	5.45 ± 0.05	60.78 ± 6.44	8.52 ± 0.48	7.81 ± 0.48

TABLE IV

CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE CP DATASET USING 0.2% OF THE AVAILABLE LABELED SAMPLES FOR TRAINING

Class	Conventional		CNN-based Methods		Transformer-based Methods			Mamba-based Methods			GPSMamba
	SVM	KNN	3DCNN	CSCANet	DBCTNet	GSCVIT	DiffFormer	3DSSMamba	IGroupSSMamba	DBMGNet	
class1	90.58 ± 1.28	87.05 ± 1.36	86.40 ± 2.17	98.63 ± 0.36	88.23 ± 2.57	97.68 ± 1.19	95.05 ± 0.78	86.50 ± 8.27	97.46 ± 0.56	98.02 ± 1.00	**98.83 ± 0.81**
class2	72.49 ± 12.09	66.21 ± 10.23	45.05 ± 27.77	73.61 ± 6.14	81.33 ± 8.82	79.14 ± 12.54	81.76 ± 7.28	30.42 ± 11.34	63.98 ± 5.36	82.11 ± 7.94	**86.66 ± 7.91**
class3	62.54 ± 8.83	23.63 ± 7.71	0.94 ± 2.31	85.32 ± 5.08	99.18 ± 2.09	87.57 ± 11.46	75.02 ± 6.63	16.10 ± 14.19	93.15 ± 3.45	92.25 ± 4.78	**92.41 ± 8.47**
class4	59.06 ± 7.16	40.95 ± 3.50	37.37 ± 7.05	71.52 ± 1.60	72.64 ± 9.85	70.21 ± 12.97	62.34 ± 4.12	56.16 ± 10.91	91.34 ± 2.61	76.31 ± 13.64	**78.08 ± 11.49**
class5	96.60 ± 2.11	95.56 ± 2.48	98.02 ± 0.53	99.78 ± 0.17	87.23 ± 5.78	95.51 ± 2.83	97.09 ± 1.38	86.04 ± 7.64	99.18 ± 0.49	95.11 ± 0.96	**99.62 ± 1.01**
class6	20.76 ± 7.97	20.56 ± 6.92	0.00 ± 0.00	1.79 ± 2.71	43.59 ± 14.56	28.51 ± 9.31	23.03 ± 12.65	0.90 ± 1.55	43.84 ± 3.33	72.95 ± 8.80	**73.32 ± 11.99**
class7	65.61 ± 4.40	42.42 ± 4.46	52.37 ± 8.26	77.63 ± 3.65	88.79 ± 4.56	81.77 ± 3.30	62.09 ± 2.39	44.25 ± 8.41	76.98 ± 2.28	83.36 ± 2.35	**85.57 ± 4.53**
class8	96.03 ± 0.64	90.32 ± 1.08	95.25 ± 0.65	98.06 ± 0.46	93.60 ± 2.41	97.52 ± 0.85	97.29 ± 0.54	84.22 ± 4.18	97.73 ± 0.28	98.21 ± 0.92	**98.54 ± 0.47**
class9	93.42 ± 3.66	92.55 ± 4.08	86.75 ± 10.06	98.55 ± 1.18	97.45 ± 2.66	95.15 ± 5.55	94.90 ± 3.65	84.96 ± 10.02	98.28 ± 0.93	98.02 ± 1.79	**97.56 ± 2.57**
class10	87.49 ± 8.20	76.82 ± 8.89	68.54 ± 4.47	98.51 ± 1.22	86.20 ± 7.14	90.91 ± 6.87	98.40 ± 1.75	80.20 ± 12.05	98.57 ± 1.08	95.84 ± 2.83	**95.96 ± 3.09**
class11	13.33 ± 14.14	12.63 ± 5.68	0.00 ± 0.00	25.74 ± 9.72	59.25 ± 11.19	34.08 ± 16.27	8.72 ± 5.82	10.52 ± 11.00	42.92 ± 4.03	56.61 ± 13.94	**59.70 ± 13.25**
class12	28.45 ± 10.37	16.45 ± 10.75	0.92 ± 1.88	56.51 ± 5.23	68.36 ± 5.65	53.56 ± 18.25	34.94 ± 10.16	13.76 ± 6.50	60.22 ± 3.44	57.48 ± 10.30	**60.02 ± 13.33**
OA(%)	88.21 ± 0.46	81.71 ± 0.43	82.57 ± 1.46	94.41 ± 0.22	89.38 ± 0.98	93.47 ± 0.62	90.75 ± 0.36	78.28 ± 4.31	94.66 ± 0.31	94.86 ± 0.28	**95.62 ± 0.72**
AA(%)	65.53 ± 2.18	55.43 ± 1.12	47.63 ± 3.25	73.80 ± 1.30	80.49 ± 1.82	75.97 ± 2.08	69.22 ± 1.88	49.50 ± 4.40	80.30 ± 0.66	83.86 ± 1.66	**84.69 ± 3.06**
Kappa(%)	84.43 ± 0.61	75.70 ± 0.58	76.88 ± 1.94	92.63 ± 0.29	86.21 ± 1.26	91.41 ± 0.82	87.76 ± 0.46	71.14 ± 5.72	92.98 ± 0.41	93.24 ± 0.36	**94.24 ± 0.94**
Training Time	**0.01 ± 0.00**	**0.01 ± 0.00**	10.58 ± 1.08	28.81 ± 0.68	15.40 ± 0.21	24.30 ± 1.48	18.04 ± 0.13	29.44 ± 0.06	118.25 ± 7.80	638.66 ± 165.04	357.88 ± 4.66
Test Time	45.53 ± 2.59	**0.27 ± 0.04**	37.79 ± 4.12	23.99 ± 0.46	6.00 ± 0.19	157.10 ± 6.19	43.84 ± 0.26	22.97 ± 0.06	218.35 ± 20.88	34.78 ± 13.61	27.43 ± 0.31

convergence speed and classification performance for all datasets.

D. Comparison With SOTAs

To verify the effectiveness of the proposed method, the GPSMamba is compared with four types of mainstream HSIC methods. The OA, AA, and Kappa coefficient are adopted as quantitative evaluation metrics. The classification results are presented in Tables I–IV, with the optimal results highlighted in boldface. The classification visualization maps are illustrated in Figs. 12–15.

1) PU Dataset: Experiments on the PU dataset are conducted using 1% of the available labeled samples as the training set. Table I summarizes the classification results on the PU dataset. GPSMamba achieves the optimal performance, with OA, AA, and the Kappa coefficient reaching 98.98%, 98.17%, and 98.65%, respectively. Compared with IGroupSS-Mamba, the three metrics are improved by 0.84%, 1.07%, and 1.12%, respectively. The performance gaps among existing hyperspectral classification methods mainly stem from differences in their feature modeling mechanisms. SVM and KNN rely only on handcrafted shallow spectral features for classification and lack the capability to model spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

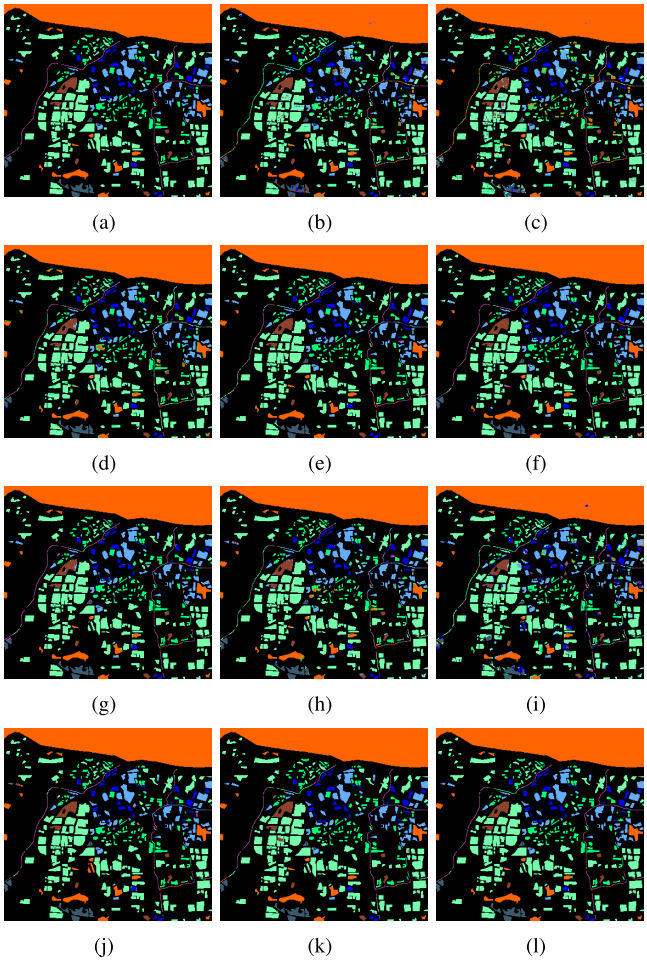


Fig. 14. Classification visualizations for the HZ dataset based on 1% labeled training samples per class. (a) Ground truth, (b) SVM, OA = 95.90%, (c) KNN, OA = 93.96%, (d) 3DCNN, OA = 94.15%, (e) CSCANet, OA = 97.72%, (f) DBCTNet, OA = 98.04%, (g) GSC-ViT, OA = 97.26%, (h) DiffFormer, OA = 94.97%, (i) 3DSS-Mamba, OA = 86.68%, (j) IGroupSS-Mamba, OA = 97.34%, (k) DBMGNet, OA=98.90%, (l) GPSMamba, OA = 98.97%.

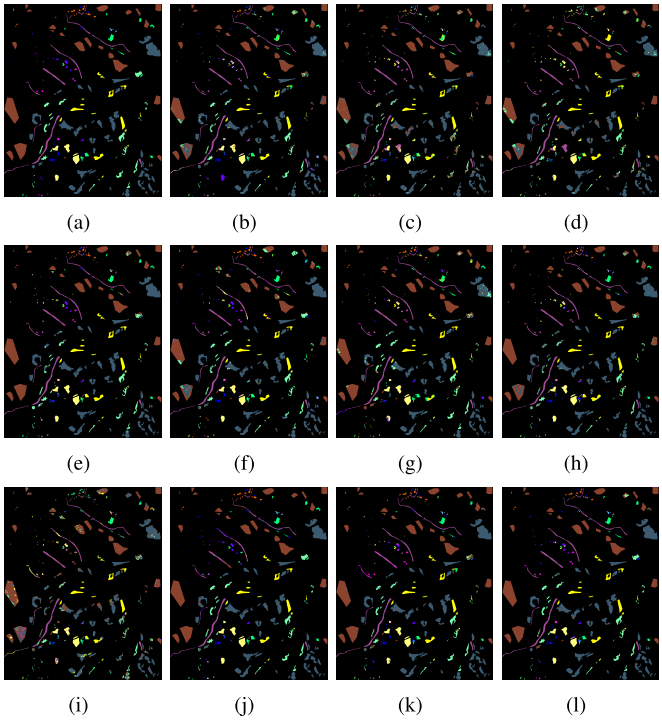


Fig. 15. Classification visualizations for the CP dataset based on 0.2% labeled training samples per class. (a) Ground truth, (b) SVM, OA = 88.21%, (c) KNN, OA = 81.71%, (d) 3DCNN, OA = 82.57%, (e) CSCANet, OA = 94.41%, (f) DBCTNet, OA = 89.38%, (g) GSC-ViT, OA = 93.47%, (h) DiffFormer, OA = 90.75%, (i) 3DSS-Mamba, OA = 78.28%, (j) IGroupSS-Mamba, OA = 94.66%, (k) DBMGNet, OA = 94.86%, (l) GPSMamba, OA = 95.62%.

sequences and joint spectral–spatial information. 3DCNN and CSCANet extract spectral–spatial features via local receptive fields and suppress partial noise, but they are incapable of modeling long-range dependencies. The DMHSA module in DiffFormer focuses only on local spectral–spatial differences and completely ignores global spectral–spatial features. DBCTNet introduces joint CNN–transformer modeling to mine global features and optimize feature extraction capacity. GSC-ViT and IGroupSS-Mamba improve model efficiency through grouping strategies, yet they break spectral continuity and result in the loss of local details. 3DSS-Mamba and DBMGNet perform global modeling for HSI based on Mamba's sequence modeling ability, but they fail to address spatial information loss during the dimensionality conversion of HSI. GPSMamba effectively alleviates the above issues through three core modules: DMC, DAM, and DSA. Specifically, DMC dynamically models global context while retaining convolutional local inductive biases, realizing dynamic extraction of global–local features. DAM mitigates spatial information fragmentation via SSS. DSA suppresses spectral confusion between similar ground objects and further enhances the discriminability of spectral features. As shown in Table I, the proposed method achieves significant improvements on multiple categories including Asphalt (C1), Trees (C4), and Bricks (C8). For Metal sheets (C5), Bitumen (C7), and Shadow (C9), GPSMamba obtains comparable performance to the state-of-the-art baselines. This phenomenon can be attributed to the low classification difficulty of these land-cover samples, where existing approaches already deliver accurate discrimination results. The visualization results in Fig. 12 further verify that GPSMamba produces classification maps with fewer misclassified pixels and sharper class boundaries.

2) SA Dataset: Experiments on the SA dataset are conducted using 1% of the available labeled samples as the training set. Table II summarizes the classification results on the SA dataset. As shown in Table II, the proposed method achieves the optimal overall performance with OA, AA, and the Kappa coefficients reaching 99.29%, 99.51%, and 99.21%, respectively, which demonstrates the remarkable effectiveness of the proposed method for HSIC tasks involving crop categories with highly similar spectral characteristics. Although methods such as DiffFormer, DBCTNet, and GSC-ViT can realize global feature modeling via the self-attention mechanism, they suffer from inadequate spatial continuity modeling, which limits their classification performance in large-scale farmland scenarios. Meanwhile,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

CSCANet adaptively extracts local texture features through multiscale receptive fields and delivers more comprehensive HSI feature representations compared with 3DCNN. In contrast, Mamba-based methods represented by 3DSS-Mamba and DBMGNet mainly rely on sequence modeling, and the grouping strategy of IGroupSS-Mamba impairs spectral features, which restricts their feature representation capacity for large-scale complex spatial–spectral features. The superior performance of GPSMamba is mainly attributed to the synergistic effect of DMC, DAM, and DSA. Specifically, DMC realizes multiscale aggregation of local features embedded with global information, DAM enhances the model's understanding of HSI spatial structures, and DSA improves the model's discriminative capability for distinct spectra, thereby boosting the discriminability of HSI features. As shown in Fig. 13, the SA dataset mainly consists of contiguous farmlands with highly similar spectral signatures among various crop types. Although CSCANet and IGroupSS-Mamba can obtain competitive overall OA values, they still suffer from extensive misclassification on individual crop categories such as Lettuce-romaine-4wk (C11) and Vineyard-vertical-trellis (C16). By contrast, GPSMamba achieves the most prominent accuracy gains on these two classes and delivers superior boundary segmentation simultaneously. For Lettuce-romaine-5wk (C12) and Lettuce-romaine-6wk (C13), the classification results of our proposed method are comparable yet slightly inferior to state-of-the-art competitors, which reveals that existing algorithms can already handle these land-cover categories effectively.

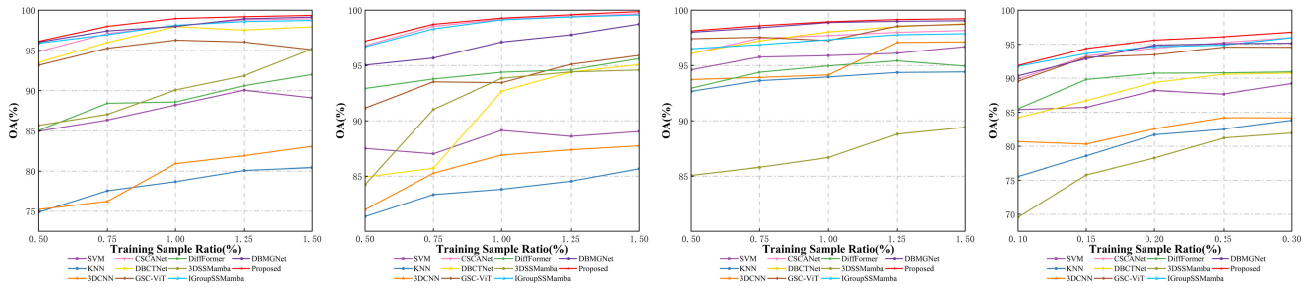
3) HZ Dataset: Experiments on the HZ dataset are conducted with 1% of the available labeled samples used as the training set, and the quantitative classification results and visualization maps are given in Table III and Fig. 14, respectively. GPSMamba achieves the optimal overall performance, with the OA, AA, and the Kappa coefficient reaching 98.97%, 95.99%, and 98.48%, respectively. Compared with the DBMGNet method, the three metrics increase by 0.07%, 0.07%, and 0.10%, respectively. 3DCNN and CSCANet rely on local feature modeling and cannot adapt to HSI with scattered ground-object distributions. DBMGNet models long-range spatial–spectral dependencies via Mamba and simultaneously leverages GCN to capture non-Euclidean spatial similarity relationships, allowing it to fit the spatial structures of scattered ground objects well. DiffFormer, DBCTNet, and GSC-ViT perform global context modeling to extract topological structures of large-area ground objects. Due to spatial fragmentation or separated spectral features of ground objects, 3DSS-Mamba and IGroupSS-Mamba easily produce semantic confusion between spectrally similar ground objects. By adopting global–local dynamic mixing, GPSMamba effectively mitigates spectral confusion and spatial fragmentation via spatial sequence modeling and spectral decoupling. Specifically, DMC captures mixed global–local features of fine-grained ground objects, DAM models the spatial continuity of HSI through direction-aware updates, and DSA reduces semantic confusion for similar classes such as water bodies and shadows, enabling precise recalibration of similar ground objects. As shown in Table III, GPSMamba achieves the most prominent optimization on Reed Field (C1), Pond (C4) and Forest (C8). In addition, it obtains the optimal recognition performance for fragmented ground objects of Rice Field-After Weeding (C6). As depicted in Fig. 14, benefiting from the above structural designs, GPSMamba produces classification maps with superior ground boundary segmentation, and its prediction results are more consistent with the ground truth labels.

4) CP Dataset: Experiments on the CP dataset use 0.2% of the available labeled samples as the training set, and the quantitative classification results and visualization maps are presented in Table IV and Fig. 15, respectively. GPSMamba still achieves optimal performance in complex coastal wetland scenarios, with OA, AA, and Kappa reaching 95.62%, 84.69%, and 94.24%, respectively. Compared with DBMGNet, OA rises by 0.76%, AA by 0.83%, and Kappa by 1.00%. The CP dataset poses great challenges, as it contains complex coastal wetland and nonwetland classes with imbalanced sample distribution and prominent spectral variability. Poor-performing models mostly lack sufficient long-range dependency modeling capacity or cannot adapt to the unique ground-object topological relationships of coastal wetlands, which leads to classification regions contaminated by cross-category noise induced by similar spectra. 3DSS-Mamba flattens 3-D sequences, which causes the loss of distinctive wetland spatial morphology and weakens its spatial–spectral adaptability to wetland scenes. In contrast, advanced benchmark methods including DBMGNet, IGroupSS-Mamba, and GSC-ViT excel at capturing richer spatial structures and topological relationships, thereby boosting classification performance. Compared with these methods, GPSMamba sustains top-tier classification capacity: it aggregates multiscale local details and global context via DMC, and separately enhances spatial structure modeling and spectral correlation learning through the DAM and DSA branches. Specifically, DMC leverages dynamic convolution kernels to fit scattered ground-object layouts, DAM captures distinctive topological patterns of coastal land covers, and DSA mines cross-dimensional spectral correlations. As shown in Table IV, GPSMamba achieves the most significant improvements for Impervious (C2) and the irregular wetland River (C5), and realizes optimal boundary fitting for Brick house (C3). Nevertheless, Bare (C4) and Lake (C6) remain challenging for classification, which is caused by variable spectral responses under different water contents and surface conditions. As illustrated in Fig. 15, the classification regions generated by GPSMamba retain clearer ground-object boundaries and category areas, and the predicted map has a better match with ground truth annotations.

E. Ablation Study

To verify the effectiveness and cooperative mechanism of the three core components (DMC, DAM, and DSA) within the proposed GPSMamba framework, ablation experiments are carried out. Starting from the complete GPSMamba model, we remove single or multiple core modules for testing across four standard HSI datasets: PU, SA, HZ, and OA are adopted uniformly as the evaluation metric.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 16. Classification performance under varying training sample proportions on four datasets. (a) PU. (b) SA. (c) HZ. (d) CP.

TABLE V

ABLATION STUDY OF THE KEY COMPONENTS OF GPSMAMBA (BEST RESULTS IN BOLD)

DMC	DSA	DAM	OA(%)			
			PU	SA	HZ	CP
✓	×	×	98.46	98.80	98.42	95.05
×	✓	×	98.21	97.80	97.72	91.81
×	×	✓	98.71	96.37	97.41	88.80
×	✓	✓	98.55	97.99	97.64	91.54
✓	×	✓	98.40	98.86	97.92	95.08
✓	✓	×	98.03	99.16	98.35	95.23
✓	✓	✓	**98.98**	**99.29**	**98.97**	**95.62**

The proposed DMC addresses the limitation that CNNs cannot capture long-range dependencies. It dynamically generates convolution kernels embedded with long-range contextual information to extract local features of HSI, thereby enabling effective global–local feature extraction for HSI. DAM executes SSS following the DPU mechanism and incorporates directional update parameters during state space updates. This alleviates spatial fragmentation caused by spatial dimension transformation and improves the model's grasp of HSI spatial structures. DSA decouples features along spatial and channel dimensions separately. It models interband spectral dependencies and intraband pixel correlations via spatial and channel attention matrices, which optimizes spectral perception and the interpretation of HSI spatial layouts. To evaluate the performance of each component, we disable one module at a time and record the resulting performance shifts. As shown in Table V, the model achieves noticeably lower accuracy when any single module is removed from the full framework.

Specifically, deactivating DMC prevents the model from explicitly extracting long-range dependencies and convolutional local inductive biases of HSI, so it fails to learn more discriminative feature representations. The OA drops by a maximum of 4.08% on the CP dataset. When DAM is disabled, the model loses the ability to preserve intact HSI spatial structures via direction-aware updates. This worsens spatial fragmentation during the transformation from 2-D images to 1-D sequences and severely weakens spatial sequence feature extraction capacity; the OA falls to 98.03% on the PU dataset. Deactivating DSA impairs the model's capacity to capture interband spectral correlations and interpret HSI spatial structures. Its power to model deep spectral features and intraband pixel correlations declines sharply, drastically reducing the discriminative capacity of both spectral and spatial features, with the OA decreasing to 97.92% on the HZ dataset. As shown in Table V, disabling DMC leads to a much larger accuracy drop than disabling DAM on the SA, HZ, and CP datasets. As DMC enhances discriminative base features for the entire network, eliminating it deprives the model of valid underlying representations. Consequently, DAM can only execute scanning on original hyperspectral inputs and fails to capture discriminative semantics of various land covers, leading to a severe decline in classification accuracy.

We also test configurations retaining only one component to validate the synergy among DMC, DAM, and DSA. When only DMC is enabled while DAM and DSA are disabled, the model merely performs multiscale global–local semantic learning. Its capacity to perceive spatial and spectral features is limited, making it unable to fit the spatial–spectral coupling properties of HSI. When only DAM is retained, the model exclusively learns spatial sequence features. Without the auxiliary global–local feature extraction and optimized spectral correlation modeling provided by DMC and DSA, it shows extremely weak adaptability to complex ground-object scenes, with the OA dropping to 88.80% on the CP dataset. When only DSA is kept, the model cannot fully capture HSI spatial structures and global semantic information, leading to classification accuracies of 98.21%, 97.80%, 97.72%, and 91.81% on the four datasets, respectively.

In summary, the GPSMamba framework integrates DMC, DAM, and DSA to tackle core challenges in HSIC. For DMC, long-range contextual information is embedded into dynamic convolution kernels to jointly extract multiscale local details and long-range dependencies of HSI. DAM efficiently extracts 2-D spatial structural features of HSI by implementing SSS under the DPU mechanism. DSA explicitly decouples HSI along spatial and channel dimensions: its channel branch models interband spectral dependencies, while its spatial branch models intraband pixel correlations, further refining spectral perception and spatial structure interpretation. The three modules work in synergy to fully exploit complementary spatial–spectral information of HSI and deliver stable, superior classification performance. The ablation results clearly prove that every core component contributes positively to the final

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

accuracy, which demonstrates the innovation and efficacy of the GPSMamba architecture for HSI feature learning.

F. Robustness Evaluation

To evaluate the robustness of the proposed DMCSC, extensive experiments are conducted on four public hyperspectral datasets (i.e., PU, SA, HZ, and CP) by setting different proportions of labeled training samples. The classification performance curves of all comparison methods are depicted in Fig. 16, where the results of DMCSC are represented by red curves. Specifically, the proportion of labeled training samples is set between 0.5% and 1.5% for the PU and SA and HZ datasets, while it ranges from 0.1% to 0.3% for the CP datasets. The experimental results show that the classification accuracy of DMCSC exhibits a continuous upward trend with the increase of the training sample ratio. Meanwhile, DMCSC outperforms other comparison methods under most sampling ratios, which validates the robustness and effectiveness of the proposed method.

V. Conclusion

This article proposes a novel GPSMamba framework for HSIC. First, DMC is designed to calculate the affinity matrix between feature tokens and regional centers, yielding spatially variant dynamic convolution kernels embedded with contextual information. It realizes joint modeling of global context and local details in HSI and captures long-range dependencies while preserving the local inductive biases of convolutions. Second, the DPU mechanism is adopted to implement SSS, enabling DAM to effectively mitigate HSI spatial structure fragmentation during feature extraction, reduce spatial information disruption induced by HSI dimensional transformation, and deepen the model's comprehension of HSI spatial structures. Additionally, the proposed DSA decouples spatial and channel dimensions. The channel branch captures interband spectral dependencies to boost the model's spectral perception capacity for HSI, while the spatial branch models intraband pixel spatial correlations. This design further optimizes multidimensional spectral perception and the interpretation of HSI spatial structures. Extensive experimental results on multiple public HSI datasets demonstrate that the proposed model outperforms existing mainstream classification methods and retains strong performance under both complex scenes and few-shot settings. Although GPSMamba achieves outstanding HSIC performance, it suffers from excessive parameter counts, high computational complexity, and low inference efficiency. Future research will focus on lightweight architecture design and computational cost optimization for the GPSMamba framework, as well as extending the framework to broader remote sensing tasks including multimodal remote sensing fusion classification and cross-scene transfer classification.

References

[1] W. Zhou, H. Guan, Z. Li, Z. Shao, and M. R. Delavar, "Remote sensing image retrieval in the past decade: Achievements, challenges, and future directions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1447–1473, 2023.

[2] L. Liang et al., "LKMA: Learnable kernel and Mamba with spatial–spectral attention fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5530914.

[3] L. Liang et al., "DBMLLA: Double-branch Mamba-like linear attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5524315.

[4] S. Zhang et al., "Multiscale spatial graph-regularized hierarchical sparse unmixing based on the framelet transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5524620.

[5] Y. Zhang, P. Duan, L. Liang, X. Kang, J. Li, and A. Plaza, "PFS3F: Probabilistic fusion of superpixel-wise and semantic-aware structural features for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 9, pp. 8723–8737, Sep. 2025.

[6] L. Zhang, M. Zhang, J. Huang, C. Zhang, F. Ye, and W. Pan, "A new approach for mineral mapping using drill-core hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[7] J. Lai, J. Wang, Y. Hou, and P. Jia, "LPP-CNN: A lightweight privacy-preserving object classification framework for military vehicle images," *IEEE Internet Things J.*, vol. 13, no. 5, pp. 8465–8481, Mar. 2026.

[8] L. C. S. Dixit, S. S. Jha, U. R. Nelakuditi, and R. R. Nidamanuri, "Military target detection by integrated spectral design and radiative transfer modeling," *IEEE Geosci. Remote Sens. Lett.*, vol. 23, pp. 1–5, 2026.

[9] H. Fu, J. Ren, A. Zhang, P. Ma, X. Xiao, and Y. Yan, "Remote sensing for peatland conservation and sustainability: Survey and analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Aug. 2025, pp. 1398–1401.

[10] X. Li, K. Liu, T. Wu, and H. Su, "Comparison of TG-1 and EO-1 hyperion in urban land cover classification," in *Proc. 7th Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Jun. 2015, pp. 1–4.

[11] H. Deborah, N. Richard, and J. Y. Hardeberg, "A comprehensive evaluation of spectral distance functions and metrics for hyperspectral image processing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3224–3234, Jun. 2015.

[12] A. Schlamm and D. Messinger, "Improved detection and clustering of hyperspectral image data by preprocessing with a Euclidean distance transformation," in *Proc. 3rd Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Jun. 2011, pp. 1–4.

[13] W. Song, S. Li, X. Kang, and K. Huang, "Hyperspectral image classification based on KNN sparse representation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 2411–2414.

[14] A. Masjedi, Y. Maghsoudi, and M. J. V. Zoej, "A novel contextual classifier based on SVM and MRF for remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4368–4371.

[15] C. Li, Y. Ma, X. Mei, C. Liu, and J. Ma, "Hyperspectral image classification with robust sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 641–645, May 2016.

[16] S. D. Fabiyi, P. Murray, J. Zabalza, and J. Ren, "Folded LDA: Extending the linear discriminant analysis algorithm for feature extraction and data reduction in hyperspectral remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12312–12331, 2021.

[17] V. Bruni, G. Monteverde, and D. Vitulano, "Heisenberg principle-inspired filters size setting in 3D CNN for hyperspectral data classification," in *Proc. 14th Workshop Hyperspectral Imag. Signal Processing: Evol. Remote Sens. (WHISPERS)*, Dec. 2024, pp. 1–5.

[18] X. Liu, A. H.-M. Ng, F. Lei, J. Ren, L. Guo, and Z. Du, "MSLKCNN: A simple and powerful multiscale large kernel CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513314.

[19] X. Wang, Z. Zheng, J. Shao, Y. Duan, and L.-J. Deng, "Adaptive rectangular convolution for remote sensing pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 17872–17881.

[20] X. Ma, M. Zhang, Z. Kan, Y. Gao, and W. Li, "A lightweight spatial–spectral deformable CNN for UAV hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5524213.

[21] T. Zhang, G. Zhang, W. Jia, Y. Zeng, M. Hou, and J. Li, "FS-CrossMamba: Dual-branch frequency-guided spatial and spectral-attention-driven spectral cross-Mamba network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5514819.

[22] S.-Y. Chen, P.-Y. Chu, K.-L. Liu, and Y.-C. Wu, "A multichannel hybrid 2D–3D-CNN for hyperspectral image classification with small training sample sizes," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540915.

[23] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021, Art. no. 5518615.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.

[24] C. Jia, X. Zhang, H. Meng, S. Xia, and L. Jiao, "CenterFormer: A center spatial–spectral attention transformer network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5523–5539, 2025.

[25] Z. Zhang, H. Yu, Z. Jiang, K. Li, D. Pan, and W. Gui, "CentralGraphFormer: Center-guided spectral–spatial graph transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5507918.

[26] E. Zhao, Y. Su, N. Qu, Y. Wang, C. Gao, and J. Zeng, "Self- and cross-attention enhanced transformer for visible and thermal infrared hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13408–13422, 2025.

[27] H. Jiang, Q. Gao, H. Chen, and B. Zhou, "Center-pixel guided dual-path multihead attention fusion transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5505918.

[28] C. Wang, Y. Guo, M. Ye, and F. Xiong, "Cross-scene diffusion-enhanced uncertainty attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5509715.

[29] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision Mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[30] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 103031–103063.

[31] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "HSI-MFormer: Integrating Mamba and transformer experts for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621916.

[32] X. Ye, "Direction-aware spatial Mamba networks for hyperspectral image classification," in *Proc. Int. Conf. Comput. Vis., Image Process. Comput. Photography (CVIP)*, Jul. 2025, pp. 277–281.

[33] W. Cao, X. Yang, Y. Zhou, F. Wang, and X. Zhou, "DTM: DCT—Transformer-Mamba architecture for improving hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5608015.

[34] L. Liang, W. Zhang, P. Xie, B. Tu, J. Li, and A. Plaza, "ARAM: A lightweight adaptive rectangular attention Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5518914.

[35] P. Xie, L. Liang, S. Zhang, Y. Zhang, J. Li, and A. Plaza, "Conv-J2SMamba: A new hybrid convolutional and joint scan merge structured Mamba network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5519515.

[36] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[37] D. Y. Fu, T. Dao, K. K. Saab, A. W. Thomas, A. Rudra, and C. Ré, "Hungry hungry hippos: Towards language modeling with state space models," 2022, *arXiv:2212.14052*.

[38] A. Rajagopal and V. Nirmala, "Convolutional gated MLP: Combining convolutions and GMLP," in *Proc. Int. Conf. Big Data, Mach. Learn.*, Appl. Cham, Switzerland: Springer, 2021, pp. 721–735.

[39] W. Li, H. Chen, Q. Liu, H. Liu, Y. Wang, and G. Gui, "Attention mechanism and depthwise separable convolution aided 3DCNN for hyperspectral remote sensing image classification," *Remote Sens.*, vol. 14, no. 9, p. 2215, May 2022.

[40] B. Zhang, Y. Chen, S. Xiong, and X. Lu, "Hyperspectral image classification via cascaded spatial cross-attention network," *IEEE Trans. Image Process.*, vol. 34, pp. 899–913, 2025.

[41] R. Xu, X.-M. Dong, W. Li, J. Peng, W. Sun, and Y. Xu, "DBCTNet: Double branch convolution-transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509915.

[42] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[43] M. Ahmad, M. Mazzara, S. Distefano, A. M. Khan, and S. L. Ullo, "DiffFormer: A differential spatial–spectral transformer for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 10419–10428, 2025.

[44] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-Mamba: 3D-spectral–spatial Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[45] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-Mamba: Interval group spatial–spectral Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817.

[46] H. Wang, P. Zhuang, X. Zhang, and J. Li, "DBMGNet: A dual-branch Mamba-GCN network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4410517.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:32:52 UTC from IEEE Xplore. Restrictions apply.